



Electric vertical take-off and landing aircraft selection for ultra-short-haul routes: fuzzy AHP approach

Slavica Dožić^a, Jovana Kuljanin^{b,*}, Danica Babić^a, Milica Kalić^a, Jordi Pons-Prats^{b,c}

^a University of Belgrade, Faculty of Transport and Traffic Engineering, 305 Vojvode Stepe, Belgrade, Serbia

^b Universitat Politècnica de Catalunya BarcelonaTech (UPC), Physics department - Aeronautical Division, Campus PMT UPC, c/ Esteve Terrades 5, Edifici C3, 08860 Castelldefels, Barcelona, Spain

^c Centre Internacional de Mètodes Numèrics a l'Enginyeria, CIMNE, Barcelona, 08034, Spain

ARTICLE INFO

Keywords:

eVTOLs

fuzzy AHP

Ultra-short-haul routes

eVTOL aircraft selection

ABSTRACT

The analysis and understanding of the environmental effects of air transport has been attracting the attention of academia and the industry for more than 20 years. It is also evident that this environmental concern has grown in recent years, with a special focus on ultra-short-haul flights. In light of a new restriction recently introduced in some EU countries related to ultra-short-haul flight bans, and the recent developments in electric propulsion technologies, the electric vertical take-off and landing aircraft (eVTOL) offers a possibility for airlines to regain these routes. Considering the route requirements, together with the performance of electric new vehicles, the main objective of this paper is to select the new eVTOL aircraft, which could be exploited instead of conventional aircraft types. The proposed approach is based on the Fuzzy Analytic Hierarchy Process (FAHP) in combination with logarithmic fuzzy preference programming (LFPP) to derive priorities from expert judgments. The analysis considers operational and performance requirements for six eVTOLs candidate, evaluates their suitability through multi-criteria decision making, and examines how sensitive the results are to changes in the weighting of criteria. To illustrate the practical potential of eVTOL adoption, two banned European routes are analyzed. The findings highlight both the opportunities and limitations of integrating eVTOLs into short-haul networks and provide directions for future research in sustainable air transport. Additionally, the FAHP-derived solution, is consistently confirmed across all stakeholder groups as the most appropriate alternative.

1. Introduction

Air transport is one of the most important catalysts of the world economy and trade, enabling a very fast movement of goods and people on a global scale. According to IATA (2024), aviation transport directly and indirectly supports the employment of 86.5 million people, contributes over \$4.1 trillion to the global Gross Domestic Product (GDP)—representing 3.9 % of the global GDP—and carries over 4.4 billion passengers in 2023, with 5 billion expected in 2024, as well as 61.4 million tonnes of cargo. Despite the economic crisis and other turmoil, global air traffic has doubled in size every 15 years since 1977 on a global level and this trend is likely to continue in the future. On the other hand, aviation activity generates a negative impact on the environment and contributes to climate change. Although the overall contribution of aviation activity to greenhouse gases (GHG) emissions is evidently lower

compared to other industry sectors, it faces serious limitations for further decarbonisation. The current focus is on technological solutions, sustainable aviation fuels (SAF) and carbon compensation mechanisms as well as some market-based mechanisms which require engagement and collaboration among countries. The Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA) introduced by ICAO is currently the most comprehensive global market-based measure which aims to ensure the low carbon growth of the international aviation sector from 2020 onward (Lyle, 2018; Maertens et al., 2020; Prussi et al., 2021). Additionally, the European Union included the aviation sector in the European Union Trading Scheme (EU-ETS), with similar ambitions (Hepburn et al., 2006; Anger and Köhler, 2010; Nava et al., 2018).

In an attempt to promote the sustainable and green transport sector and combat climate change, some countries of the European Union went one step further by preventing ultra-short-haul flights between cities

* Corresponding author.

E-mail addresses: s.dozic@sf.bg.ac.rs (S. Dožić), jovana.kuljanin@upc.edu (J. Kuljanin), d.babic@sf.bg.ac.rs (D. Babić), m.kalic@sf.bg.ac.rs (M. Kalić), jordi.pons-prats@upc.edu (J. Pons-Prats).

<https://doi.org/10.1016/j.jairtraman.2025.102909>

Received 31 December 2024; Received in revised form 22 August 2025; Accepted 16 September 2025

Available online 18 September 2025

0969-6997/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

that are linked by rail, with the journey duration of less than approximately 3 h (e.g., France, Belgium, and Austria). The justification of such an initiative resides in the fact that ultra-short-haul flights generate twice as many emissions per seat-kilometre than long-haul flights, and they can be replaced with less polluting high-speed rail (HSR) options (Grimme and Jung, 2018). Recently, many national authorities at the national and EU levels started to consider the idea of banning ultra-short-haul flights and replacing them with HSR options including countries such as the Netherlands, Spain and Germany. Also, the attention has begun to turn to private air travel, starting a discussion of extending the ban toward personal air travel. According to Sun et al. (2022), these types of flights are extremely inefficient, and have not been scrutinized, while commercial airlines have been.

Although HSR is a very good alternative for many banned ultra-short routes in Europe where it exists, it is not a perfect substitute for all city pairs or passenger needs. Replacing flights with rail services raises some important questions. Does the rail network have spare capacity for extra passengers? If not, how quickly the needed new capacity can be added (new lines, tracks, vehicles)? Will this extra rail capacity be prioritized over other rail demands (commuter and freight)? The main problem for the European rail network is that it is facing capacity challenges, particularly on certain lines and during peak hours (e.g. Paris–Lyon–Marseille, Brussels–Amsterdam, Frankfurt–Cologne) (Rico and Heyl, 2025). Many HSR lines share tracks with slower regional or freight trains, so scheduling extra services can be complex. The same applies for rail timetables where acquiring slots for new trains may not exist without displacing others. An additional problem can be that many travelers still do not see the train as an attractive mode of transport (Witlox et al., 2022). It is notable that, although rail transport is widely recognized for its safety and low environmental impact, rail services still represent only a relatively small proportion of overall travel demand (Eurostat, 2025). This can be explained by the fact that airlines better meet travelers' expectations for price, convenience, and comfort, while rail services remain fragmented and nationally focused (Witlox et al., 2022). Airlines put the customer first with competitive prices, easy booking, and seamless service, whereas many rail operators focus on domestic operations, not on optimizing international or connecting travels. Another reason travelers avoid rail services with connections is unclear passenger rights. Unlike airlines, rail operators often lack clear rules for rebooking or compensation if a delay causes a missed connection (Witlox et al., 2022). At the beginning of 2023, many representatives of the aviation industry invoked the EU's freedom of movement rights to oppose environmental restrictions on short-haul flights under 500 km. Industry would like to avoid that this initial ban affects other flights in the future, while IATA, through its General Director Mr Willie Walsh, also claims this action would reduce only 4 % of CO₂ emissions in Europe (Euractiv, 2023) affecting around 24.1 % of European flights (EUROCONTROL, 2021). Furthermore, even if it were feasible to completely replace all flights in this category with rail, it would still fail to achieve the full 3.8 % cap on total CO₂ emission savings, as passengers are likely to adopt environmentally friendly transport options only if cost and speed are comparable (European Commission, 2020). A similar conclusion was reached by Dobruszkes et al. (2022), who argued that targeting shorter flights would contribute minimally to reducing aviation's climate impact and that greater efforts are required to address emissions from longer flights.

In this paper, we considered an alternative solution that could be a compromise between returning air transport to banned routes and carbon-cutting measures. It is an introduction of new electrical vertical take-off and landing (eVTOL) vehicles, with less-polluting characteristics, that will be able to cover a different set of origin and destination pairs and potentially enable the feasibility of the previously banned routes. In its Digital European Sky report, SRIA (2020) already stipulated that the application of the new eVTOLs leveraged by the Urban Air Mobility (UAM) concept shall not increase the environmental footprint of the air transportation system. However, the same document claims

that UAM will be gradually evolved over the next thirty years, from piloted to autonomous operations. According to the projection performed by HDI (2022), it is anticipated that between 2025 and 2030 small hybrid-electric aircraft may enter the market of commercial segments. Given the range of these new vehicles will to some extent coincide with the one provided by the traditional business jet planes, one could anticipate the replacement of the former ones on specific routes. Therefore, the selection of adequate eVTOLs for the airlines' fleet will be of paramount importance in the upcoming years, as it will directly affect the competitive positions and sustainability of airlines. Importantly, this paper addresses the problem within the broader Advanced Air Mobility (AAM) context, where UAM represents only a subset of AAM (EASA, 2021). As highlighted by Mallela et al. (2023), AAM encompasses not only use cases in urban environments but also short-haul and regional operations, thus providing the theoretical foundation for considering eVTOL deployment on banned ultra-short-haul routes alongside their potential intra-urban applications.

The eVTOLs may vary in size and range, defining in this way the potential exploitation. The majority of papers consider on-demand UAM taking into account the possibilities of commutes to/from work or other routine trips around a city, airport shuttles to provide faster connectivity for airline passengers to/from existing airports (from close to their origin or to near their final destination), end-to-end city transfer allowing passengers to bypass city traffic and quickly move from one side of town to another, and "metro-like" services to connect passengers to other existing forms of mass transit. A review of the current technology (including electric power systems) and research as well as the advantages and challenges of this kind of mobility in the future are given by Vieira et al. (2019), Polaczyk et al. (2019) and Barzkar and Ghassemi (2020). However, the further development of eVTOLs with larger capacities and their corresponding certification will foster the opportunity for broader applications aiming to be a dominant means of transport in the current short-haul routes across Europe.

The main objective of this paper is to select the new eVTOL aircraft which could be exploited instead of conventional aircraft types assuming the given route length. This problem belongs to the group of multi-criteria problems that are typically performed by decision-makers. Planners usually have to make decisions under uncertainty, incorporating the vagueness of human thinking. Therefore, we proposed the Fuzzy Analytic Hierarchy Process (FAHP) based approach. To derive crisp priorities from fuzzy pairwise comparison matrices, a logarithmic fuzzy preference programming (LFPP) method is employed (Dožić et al., 2018). The paper is divided into six sections. After a brief Introduction given in Section 1, the paper continues with the literature review presented in Section 2. Section 3 explains the methodology used to address the above-mentioned problem of aircraft selection. Section 4 presents the selection of an eVTOL that fits the airline's needs best. Section 5 provides a sensitivity analysis of the solution for the importance of criteria changes and evaluate potential differences in different stakeholders' prioritization of criteria and rankings of eVTOL alternatives from. Additionally, air travel demand and frequency are quantified to illustrate the possibility of eVTOLs operating banned European routes. Concluding remarks and future research directions are given in the last section.

2. Literature review

2.1. Literature review - new shapes of mobility

As already mentioned, Urban Air Mobility (UAM) is a new concept in transportation that is foreseen to partially tackle the problem of large congestion in urban and regional environments. Thus, very large metropolitan areas (e.g., Mexico City, Los Angeles, London, etc.) are deemed suitable candidates for the introduction of these new vehicles due to their persistent problems of long commuting times, pollution and congestion. UAM is aimed to cover a variety of possible missions offering

the service to a wide spectrum of potential markets and customers. For instance, in its Elevating the future of mobility report, [Deloitte \(2019\)](#) argued that new types of aircraft will revolutionize inter- and intra-city mobility. While the former one aims at providing on-demand air transportation within densely populated cities and mainly connecting the city centre with the airport, the latter one is focused on offering the service between cities and suburban areas. The study of [KPMG \(2019\)](#) envisions the gradual evolution of UAM services in three stages, starting with intra-city airport shuttles in the 2030s, followed by on-demand and scheduled air-taxi services (with driver and autonomous). In addition, in some cities such as New York, a similar concept has been already proven by the application of on-demand helicopter service. BLADE offers the transport of passengers from several locations in Manhattan to one of the three main airports in New York City (John F. Kennedy International Airport - JFK, LaGuardia Airport - LGA, and Newark Liberty International Airport - EWR). In a similar vein, in collaboration with HeliFlite4, Uber started offering flights from Manhattan to JFK airport during the weekdays from 1 p.m. to 5 p.m.

However, when dealing with different UAM missions, one cannot neglect the aspect of eVTOLs operational requirements. Summarizing available data on the UAM missions, the National Aeronautics and Space Administration ([Patterson et al., 2018](#)) found that the majority of the proposed missions have ranges of less than 100 nautical miles (185.2 km), with the capability of transporting nine or fewer passengers. On the other hand, the longest-range missions (those at 174 nautical miles or 322 km) aim to connect the centre of the metropolitan areas and its surrounding suburban/rural areas or two suburbs, with a maximum capacity of 9 passengers. In this regard, ([Patterson et al., 2012](#)) attempts to relate the range-payload requirements of the respective UAM mission with the most suitable reference aircraft. For each of the three longest-range missions proposed, ([Patterson et al., 2018](#)) defined the operational characteristics that eVTOL needs to meet in order to safely operate different flight phases. Based on the aggregation of data from twenty-eight metropolitan areas, ([Patterson et al., 2018](#)) defined a typical "short-range, large-payload mission" characterised by a range of 37.5 nautical miles (69.5 km) and the payload which weight may vary depending on the seat number, i.e., 11 lb in 6-seat aircraft, and 81 lb in 4-seat aircraft. Nevertheless, all these missions assume that the capacity of the eVTOL vehicles will not exceed nine passengers on-board which might be insufficient to mitigate the severe traffic congestion currently perceived in large metropolitan areas. [NASA \(2021\)](#) recently updated these assumptions by introducing the Advanced Air Mobility (AAM) Mission Concept, which highlights vehicle operations, airspace design, and community integration as the main pillars for enabling sustainable and scalable medium-density AAM operations. Building on this foundation, the [FAA \(2023a\)](#) released AAM Implementation Plan which aims to further elaborate and guide implementation efforts and mature as the FAA works with stakeholders to refine and execute AAM implementation strategies. Additionally, the [AirMOUR project \(2023\)](#) provides a comprehensive framework for integrating UAM into urban environments, highlighting the need for coordinated planning across urban design, aviation safety, public acceptance, and regulatory processes to ensure successful deployment and operation.

It is also worth noting that adding additional infrastructure to the current transport network (e.g., new major roads) cannot alleviate the congestion, but may even exacerbate the entire network performance according to the very known principle of road congestion. Thus, the provision of sufficient UAM capacity would be essential for the successful implementation of these new modes of transport in large metropolitan areas. Some authors (([Moore et al., 2013](#)); ([Patterson et al., 2012](#))) claim that the future UAM environment will consist of easily-rentable and highly-automated eVTOL aircraft, while some others go even further arguing that the future UAM concept may serve as aerial mass transit with the eVTOLs' capacity of 6-, 15-, and 30-passengers. Some of these hypotheses have been already close to realisation, as Joby recently completed its first piloted flight between two California

airports, Marina Municipal Airport (OAR) to Monterey Regional Airport (MRJ), a major step toward commercial readiness, especially in FAA-controlled airspace ([The Wall Street Journal, 2025](#)). Similarly, the shift from pilot demonstration to near-term deployment is performed by the Volocopter non-passenger test flight that was conducted on the final day of the Olympic Games held in Paris in 2024 ([Volocopter, 2024](#)). These flights were accompanied by regulatory advances captured in the FAA's UAM ConOps v2 ([FAA, 2023b](#)) and EASA's vertiport design specifications ([EASA, 2022](#)).

However, the operating cost associated with these new vehicle types remains one of the prevailing factors which may slow down the commercialisation of the entire concept. First of all, the acquisition of these new vehicles requires extensive financial sources, with the price of a typical flying car from \$200,000 to \$600,000, which is higher compared to the cost of conventional or fully autonomous vehicles ([Wadud, 2017](#)). Moreover, the range for the operating cost varies substantially depending on the available reports. For instance, Uber Elevate estimates that at the launch of their on-demand air taxi, the service costs will be \$5.73 per pax-mile but will decrease in the near term to \$1.84 by increasing utilization through ride-hailing ([Holden, 2018](#)). On the other hand, some other reports such as those provided by NASA are not that optimistic forecasting the costs of a five-seat eVTOL at \$6.2 in the near term, while in the long term, with further advancement in technology, the cost may diminish to \$3.75/pax-mile (([Reiche et al., 2018](#))). For the sake of comparison, it is worth noting that the eVTOL operations are likely to be operated at costs that are lower than the current helicopter service whose cost estimation ranges between \$6 and \$8 per seat-mile ([Johnson et al., 2020](#)).

In addition to the extensive cost related to the UAM operations, one of the main hurdles to the introduction of the new vehicles is undoubtedly public acceptance. According to the market study performed by [Yedavalli and Mooberry \(2019\)](#), the five top factors related to public acceptance are structured around the safety of the people on the ground, the type and level of noise emitted by the vehicles, time of day during which the vehicles travel and the altitude at which they fly. In a similar vein, [Eker et al. \(2020\)](#) underlined the two fundamental pillars of public acceptance: (i) the anticipated benefits and concerns arising from the future use of flying cars; and (ii) the public adoption of flying cars, as expressed through their acquisition or use by the commuting population. In the first study, [Eker et al. \(2019\)](#) used statistical analysis to gain insight into how individual-specific characteristics may affect public perception, proposing a set of recommendations and guidelines to policy-makers and other involved stakeholders. In order to complement the previous study, [Eker et al. \(2020\)](#) conducted an online survey of 692 individuals with the aim to identify the perceptual patterns of individuals towards travel time, cost and environmental benefits, as well as towards challenges arising from key flying cars operational characteristics. The finding outlines that a number of socio-demographic, behavioural, and attitudinal characteristics affect respondents' expectations and concerns about the adoption and implementation of flying cars. In a similar vein, [EASA \(2021\)](#) highlights the importance of public acceptance, particularly regarding noise, safety, and equitable access, as key considerations for the successful introduction of urban air mobility services. Complementing this regulatory perspective, [Riza et al. \(2024\)](#) conducted a scenario-based study examining user acceptance of air taxis in different urban settings. Their findings show that integrating air taxis with existing public transportation networks significantly improves user acceptance, while standalone air taxi services may face more resistance from urban populations.

The European airline market is specific since it is divided by countries, which may have different laws, and which may additionally increase complexity. Also, some European countries have already banned some routes to reduce the negative influence on congestion and pollution. The air travel demand is forced to be shifted to rail transport since it is more environmentally friendly. Bearing in mind all the above-mentioned, eVTOLs appear as reasonable solutions for airlines to

reduce negative impact and to preserve their presence in the airline market.

2.2. Fleet selection problem

The selection of the appropriate aircraft types and their size is one of the most demanding problems in the strategic planning of airline operations. The airline typically endeavours to match its capacity with the demand in the given prevailing market conditions. The outcome of this undertaking will have a direct impact on airline operating costs and consequently on its final financial performance. The process of aircraft selection is very complex for an airline as it requires a multidisciplinary approach employing experts from various fields. Typically, the decision on the acquisition needs to be assessed from different perspectives including economic factors, technological performance, level of service that needs to be offered to passengers, environmental impact and others (Dožić, 2019). Bearing in mind that all these factors contribute to the final decision on aircraft acquisition, it is not surprising that a large number of studies found in the relevant literature treat the aircraft selection and evaluation issues as a multi-attribute problem employing different multi-criteria decision-making (MCDM) methods. Not surprisingly, the decision requires a trade-off between different, very often opposing criteria, which need to be taken into account during this demanding process. In the pioneering study of See et al. (2004), the authors used a multi-criteria approach to launch a homogenous fleet that will operate the routes between major cities of Asia Pacific and the U.S. relying on three criteria: speed, range and number of passengers. Even though the study neglected other important factors such as costs and environmental issues, its particular contribution resides in the application of hypothetical equivalents and inequivalents methods. Yeh and Chang (2009) combined multiattribute value theory (MAVT), a simplification of Multiattribute Utility Theory (MAUT) (Keeney and Raiffa, 1993) and Technique for Order of Preference by Similarity to Ideal Solution - TOPSIS (Hwang and Yoon, 1981; Yoon, 1987; Hwang et al., 1993) to select appropriate aircraft. For this purpose, the authors identified three criteria (i.e., technological advances, social responsibility and economic efficiency), eventually resulting in eleven sub-criteria. For each of these criteria, the performance of the aircraft is evaluated by using the fuzzy rating. In an attempt to select appropriate aircraft for regional charter flights in Brazil, Gomes et al. (2014) used the NAIAD method (Novel Approach to Imprecise Assessment and Decision Environments) which is an intrinsically stochastic approach. The authors performed the aircraft evaluation process by creating a comprehensive set of twelve sub-criteria covering three broad categories, i.e., financial, logistics and quality. Moreover, Sun et al. (2011) aimed to select the aircraft for a hypothetical airline allowing some robustness in the process. The robustness is considered to efficiently deal with uncertainties stemming from fluctuations in fuel prices and variations in load factors.

Among a plethora of MCDM techniques applied in the aircraft selection process, Analytic Hierarchy Process (AHP) is widely used due to its distinctive advantages over other methods. In this regard, Bruno et al. (2015) proposed a hybrid approach that combines the strengths of the two methodologies - AHP and Fuzzy Sets. Dožić and Kalić (2014) employed the AHP approach to address the problem of aircraft type selection for regional routes. Using the same example, the authors (Dožić and Kalić, 2015a) extended the previous work by comparing the results obtained by the AHP method with those derived by the Even swap method. Moreover, in the subsequent study, the same group of authors (Dožić and Kalić, 2015b) proposed a comprehensive methodological framework for fleet planning. In a similar vein, in order to tackle the nonlinear relationship between main criteria and sub-criteria in the process of aircraft selection for Turkish airlines, Ozdemir and Basligil (2016) applied the fuzzy analytic network process (ANP) and Choquet integral methods. Finally, the results of both algorithms are compared to those obtained by using fuzzy AHP. The authors used a total number of

ten sub-criteria, structured around three main ones - costs, physical attributes and time. The method used in the recent study Dožić et al. (2018) presents a particular inspiration for the analysis performed in this paper as it allows fuzzy pairwise comparison of criteria in the case of regional aircraft selection problem. The authors used a logarithmic fuzzy preference programming (LFPP) method to obtain crisp priorities. In addition to the aforementioned studies addressing certain relevant aspects of aviation, numerous recent applications of the fuzzy-AHP method have emerged in areas such as environmental concerns (Gupta and Gupta, 2024; Dawar et al., 2025), customer behavior (Chopra et al., 2025; Preeti et al., 2024), and service quality (Sama et al., 2021). These works offer valuable methodological insights for the present eVTOL aircraft selection challenge, as they tackle closely related decision-making challenges.

Based on the previous studies, one could draw a conclusion that depending on the contextual settings and market conditions, the preferences of the airlines towards different criteria may substantially differ during the process of aircraft selection (Clark, 2007). The earlier studies typically encompass the technological factors that are considered in all previously mentioned papers. This group of factors refers to aircraft and airport characteristics, as well as aircraft performances. The second group of attributes is related to the economic aspects and is widely used in most of the studies. On the other hand, planners may struggle to obtain these factors or to precisely express them as they are given in monetary terms. Additionally, some of this data is confidential which may impose specific obstacles during the model specification. The third group aims to address the factors which reflect the level of service offered to passengers. These factors are particularly important in the era of a highly competitive market, in which airlines compete in aspects other than ticket price (e.g., comfort, luggage, safety, etc.). In recent studies (Wang et al., 2023; Güntut and Gökdalay, 2023), the environmental aspects are taking a prominent role in the fleet planning process as there is high pressure on the airline industry to reduce the generation of greenhouse gas emissions. Finally, other factors which are not classified in one of the four mentioned groups, include various criteria related to time, payment, commonality and robustness.

Bearing in mind that UAM is a novel concept leveraged by the development of eVTOLs, it is still very vague which strategies will be adopted by the airlines during the eVTOLs acquisition process. The aircraft is a very expensive asset, and acquiring new eVTOLs will impose substantial costs on airlines. Therefore, the selection of the criteria will pose a particular challenge for both the airline planners and the academia. Undoubtedly, using an all-electric fleet will contribute to the reduction of greenhouse gas emissions through the design of electric propulsion systems and batteries as well as trajectory optimisation and automation (Shihab et al., 2019; Kasliwal et al., 2019). As previously discussed, the initial assessment of the vehicles will be based on their technical specifications (e.g., range, battery lifetime, number of seats) since the cost consideration will be still unknown for most of the vehicle prototypes. The study attempts to apply the MCDM technique and select the most relevant criteria which are publicly available in order to facilitate the process of eVTOL aircraft selection on ultra-short-haul routes across Europe. As already mentioned, these routes are the subject of the new regulations of the European Commission on their ban and replacement by rail.

3. Methodology

The methodology applied to the eVTOL selection is similar to the one proposed by Dožić et al. (2018), which proved to be very useful for solving the aircraft selection problems. The fact that the data related to the new vehicles, such as eVTOLs, are not available, gives added value to the FAHP methodology, which is suitable for solving problems that involve uncertainty. It is given by Fig. 1 and consists of six steps: 1) criteria selection, 2) definition of the set of alternatives, 3) expert feedback on criteria ranking, 4) fuzzy pairwise comparison matrices



Fig. 1. The steps of the methodology proposed.

design, 5) application of FAHP and LFPP to derive crisp priorities from fuzzy pairwise comparison matrices, and 6) concordance, correlation and compatibility measures.

The sub-section to follow will describe each step of the methodology in more detail.

3.1. Criteria selection

Defining appropriate criteria is essential for making informed decisions about selecting the most suitable alternative from the available set of eVTOLs. Given that electric aircraft represent an emerging technology within the aviation industry, and considering that they are still in the development and prototype-testing phases, identifying relevant criteria has proven to be a complex challenge. The objective was to establish meaningful criteria despite the limited availability of current data.

One of the most comprehensive resources for eVTOL aircraft information is the (Electrical, 2024), which offers an extensive database categorizing various types of vehicles, such as vectored thrust, lift + cruise, and wingless (multicopter) designs. The *eVTOL Aircraft Directory* includes vehicles at different stages of technological readiness, ranging from concept designs to full-scale prototypes. Typically, the provided information focuses on technical specifications due to a lack of any other operational metrics. For example, the study of Bridgelall et al. (2023) proposed an aeronautically informed propulsion efficiency index (PEX) derived by the combination of the independent parameters such as range, payload ratio, and aspect ratio to compare the performance of eVTOL designs. However, criteria related to operational performance or exploitation costs are not available, as electric aircraft remain a forward-looking concept aligned with the EU's future-oriented goals.

Therefore, four criteria are selected, to cover technological aspects which reflect the level of service offered to passengers and environmental aspects (Fig. 2). The chosen criteria are *Seat capacity*, *Maximal Take-Off Mass (MTOM)*, *Maximum payload*, and *Range*. *Seat capacity* represents a measure of offered capacity. The larger the capacity, the greater possibility to make a profit. *MTOM* (measured in kg) is the main unit for the calculation of airport and navigation fees. *Maximum payload* indicates how much space for passengers with their baggage is available on the flight, and it can be considered as a measure of level of service. *Range* (measured in km) defines the aircraft's ability to fly a certain distance without charging batteries. Depending on the maximum range, the set of routes that can be served is determined, but as it is previously mentioned, the main focus is on the ultra-short-haul routes, in order to re-establish the air transport of passengers on the banned routes. Three of four selected criteria (*Seat capacity*, *Maximum payload*, *Range*) should be maximised, which means that the larger values of criteria are better for the airline. The only criterion which should be minimised is the *MTOM*, since all costs are anticipated to be calculated based on it.

3.2. Defining the set of alternatives

To define the set of alternatives, comprehensive research was conducted. Since the goal is to investigate the ultra-short-haul routes where air traffic is banned and replaced with rail, the authors decided to consider only electric aircraft with passenger capacity larger than two seats. This decision is supported by the fact that air transport is more carbon intensive than electric rail transport (European Environment Agency, 2021)) and that this difference could be softened by eVTOLs employment. As it has already been mentioned, eVTOLs are mostly in the development and prototype-testing phase. The set of alternatives is limited by the available data for selected criteria; therefore, it includes six electric aircraft with the desired capacity (more than two seats) as follows: Joby S4, Lilium Jet, Terafugia TF-2 Lift + Push, Alice, Jaunt, and Embraer Flexcraft.

3.3. Experts' feedback on criteria ranking

The input from experts capable of credibly assessing the alternatives across the criteria defined in the previous subsection is a crucial component of the decision-making process. To ensure reliable results, the authors conducted a survey targeting individuals with substantial expertise in the field of UAM.

The survey was structured into three main sections. The first section gathered information about the type of organization where the participant is employed and included a self-assessment of their level of expertise in UAM (on a scale from 1 to 5, where 1 represented very poor expertise and 5 indicated very good). The second section formed the core of the survey, focusing on pairwise comparisons of the previously defined criteria. This method facilitated the prioritization and relative weighting of the criteria. The third section explored expert opinions on specific topics, such as additional criteria to consider when evaluating eVTOLs, the anticipated timeline for airlines to introduce eVTOL vehicles into their fleets, and the primary obstacles to their implementation.

The survey was distributed using Google Forms. The link was shared via the authors' LinkedIn profiles and distributed directly to the email addresses of well-established experts in the field. To maximize the response rate, the survey remained open for a period of two months. Finally, a total of 13 responses were collected and validated prior to their inclusion in the analysis. In emerging fields such as eVTOLs and urban air mobility, the pool of qualified experts is relatively small and difficult to approach. To ensure quality, respondents were selected based on their experience in UAM, air transport, or transport systems analysis, and their self-assessed expertise levels were collected to validate their familiarity with the topic.

As illustrated in Fig. 3, feedback was collected from experts representing a diverse range of organizations, including airlines, Air Navigation Service Providers (ANSPs), academia, Original Equipment Manufacturers (OEMs), private companies, and other relevant entities. The respondents self-assessed their expertise with an average rating of

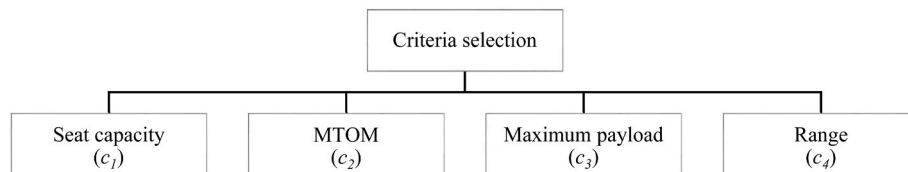


Fig. 2. Selected criteria.

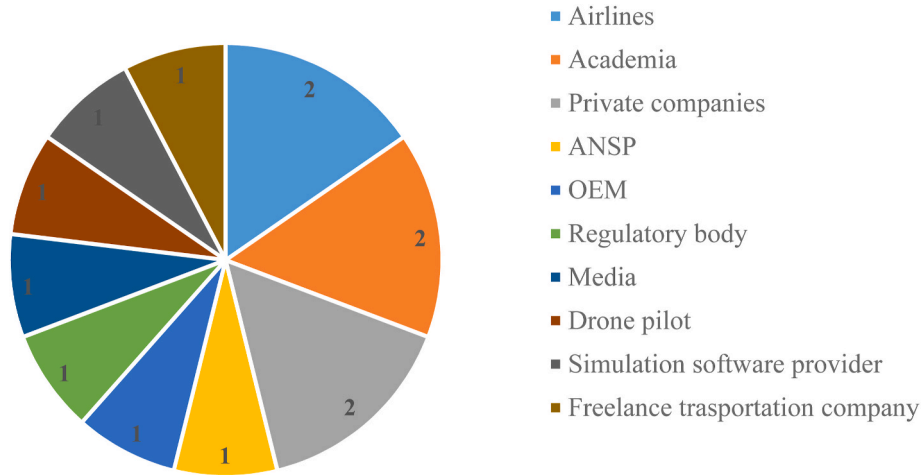


Fig. 3. Experts' affiliation.

3.3. This diversity in organizational representation allowed the study to incorporate feedback from a variety of perspectives, enriching the analysis with insights from different domains of the UAM ecosystem.

The respondents are further categorized into seven stakeholder groups: manufacturers, private companies, airlines, academia, regulatory bodies, air navigation service providers (ANSPs), and drone pilots. This classification allows for the identification of differing perspectives in the decision-making process, recognizing that such decisions are typically influenced by decision-makers' prior professional experience and educational background.

3.4. Fuzzy pairwise comparison matrices design

Table 1 presents the conversion of the linguistic scale into the triangular fuzzy scale.

Fuzzy judgements enable reflection of the vagueness and imprecision of human thought related to the problem considered. Fuzzy ratio “about a_{ij} ” represented by a fuzzy number $\tilde{a}_{ij}$ can approximate the exact numerical ratio when comparing two criteria, i and j . Fuzzy number provides information that criterion i is between l_{ij} and u_{ij} times as important as criterion j with m_{ij} being the most likely times. Based on all previously stated, (1) represents the fuzzy pairwise comparison matrix:

$$\tilde{A} = (\tilde{a}_{ij})_{n \times n} = \begin{bmatrix} (1, 1, 1) & (l_{12}, m_{12}, u_{12}) & \dots & (l_{1n}, m_{1n}, u_{1n}) \\ (l_{21}, m_{21}, u_{21}) & (1, 1, 1) & \dots & (l_{2n}, m_{2n}, u_{2n}) \\ \vdots & \vdots & \ddots & \vdots \\ (l_{n1}, m_{n1}, u_{n1}) & (l_{n2}, m_{n2}, u_{n2}) & \dots & (1, 1, 1) \end{bmatrix} \quad (1)$$

As in the pairwise comparison matrix in the case of exact numerical ratio, the elements of the matrix $\tilde{A}$ satisfy the reciprocal property, i.e. $l_{ij} = 1/u_{ji}$, $m_{ij} = 1/m_{ji}$, $u_{ij} = 1/l_{ji}$ and $0 < l_{ij} \leq m_{ij} \leq u_{ij}$ for all $i, j = 1, \dots, n; j \neq i$.

In the first step, the fuzzy pairwise comparison matrix is designed based on the data collected and the expert opinions. The average values are calculated for all answers obtained by experts, and after that these average values are transformed into the closest fuzzy numbers according

Table 1
Triangular fuzzy conversion scale for pair of elements i and j .

Linguistic scale	Triangular fuzzy scale	Triangular fuzzy reciprocal scale
Just equal	(1, 1, 1)	(1, 1, 1)
Equally important	(1/2, 1, 3/2)	(2/3, 1, 2)
Weakly important	(1, 3/2, 2)	(1/2, 2/3, 1)
Strongly important	(3/2, 2, 5/2)	(2/5, 1/2, 2/3)
Very strongly important	(2, 5/2, 3)	(1/3, 2/5, 1/2)
Extremely preferred	(5/2, 3, 7/2)	(2/7, 1/3, 2/5)

to the linguistic scale shown in Tables 1 and in the second step. Finally, after stating that the matrix is consistent (Consistency Ratio is below the threshold of 0.1, $CR = 0.045$), it is presented in Table 2. It should be mentioned that a few different methods were applied to calculate the values that will be transformed into the fuzzy numbers, but only the average values enable creation of a consistent matrix.

3.5. The logarithmic fuzzy preference programming (LFPP) method

Considering different methods to compute crisp priority vectors from the fuzzy pairwise comparison matrix, the logarithmic fuzzy preference programming method (Wang and Chin, 2011) is chosen to be applied. The logarithm values of fuzzy judgment $\tilde{a}_{ij}$ from matrix $\tilde{A}$ can be approximated by equation (2):

$$\ln \tilde{a}_{ij} \approx (\ln l_{ij}, \ln m_{ij}, \ln u_{ij}), i, j = 1, \dots, n. \quad (2)$$

The logarithm of a triangular fuzzy judgment $\tilde{a}_{ij}$ can be seen as an approximate triangular fuzzy number, whose membership function can be defined by (3):

$$\mu_{ij} \left(\ln \left(\frac{w_i}{w_j} \right) \right) = \begin{cases} \frac{\ln(w_i/w_j) - \ln l_{ij}}{\ln m_{ij} - \ln l_{ij}}, \ln \left(\frac{w_i}{w_j} \right) \leq \ln m_{ij} \\ \frac{\ln u_{ij} - \ln(w_i/w_j)}{\ln u_{ij} - \ln m_{ij}}, \ln \left(\frac{w_i}{w_j} \right) \geq \ln m_{ij} \end{cases} \quad (3)$$

where $\mu_{ij}(\ln(w_i/w_j))$ is the membership degree of $\ln(w_i/w_j)$ belonging to the approximate triangular fuzzy judgment $\tilde{a}_{ij} = (\ln l_{ij}, \ln m_{ij}, \ln u_{ij})$, and w_i are crisp values of the priority vector. A crisp priority vector would be found to maximize the minimum membership degree $\lambda = \min\{\mu_{ij}(\ln(w_i/w_j)) | i = 1, \dots, n-1; j = i+1, \dots, n\}$. Finally, the resultant model can be defined by (4) or (5):

Table 2

Fuzzy pairwise comparison matrix of selection criteria c_i with respect to the decision goal.

	Seat capacity (c_1)	MTOW (c_2)	Maximum payload (c_3)	Range (c_4)
Seat capacity (c_1)	(1, 1, 1)	(1/2, 1, 3/2)	(1, 3/2, 2)	(1, 3/2, 2)
MTOW (c_2)	(2/3, 1, 2)	(1, 1, 1)	(1/2, 1, 3/2)	(1, 3/2, 2)
Maximum payload (c_3)	(1/2, 2/3, 1)	(2/3, 1, 2)	(1, 1, 1)	(1, 3/2, 2)
Range (c_4)	(1/2, 2/3, 1)	(1/2, 2/3, 1)	(1/2, 2/3, 1)	(1, 1, 1)

$$\begin{aligned} & \max \lambda \\ & \text{Subject to } \begin{cases} \mu_{ij}(\ln(w_i/w_j)) \geq \lambda, i = 1, \dots, n-1; j = i+1, \dots, n \\ w_i \geq 0, i = 1, \dots, n \end{cases} \end{aligned} \quad (4)$$

$$\begin{aligned} & \max 1 - \lambda \\ & \text{Subject to } \begin{cases} \ln w_i - \ln w_j - \lambda \ln(m_{ij}/l_{ij}) \geq \ln l_{ij}, i = 1, \dots, n-1; j = i+1, \dots, n \\ -\ln w_i + \ln w_j - \lambda \ln(u_{ij}/m_{ij}) \geq -\ln u_{ij}, i = 1, \dots, n-1; j = i+1, \dots, n \\ w_i \geq 0, i = 1, \dots, n \end{cases} \end{aligned} \quad (5)$$

The non-negative deviation variables δ_{ij} and η_{ij} ($i = 1, \dots, n-1; j = i+1, \dots, n$) are introduced to avoid membership degree λ from taking a negative value while satisfying the inequalities (6) and (7):

$$\ln w_i - \ln w_j - \lambda \ln(m_{ij}/l_{ij}) + \delta_{ij} \geq \ln l_{ij}, i = 1, \dots, n-1; j = i+1, \dots, n, \quad (6)$$

$$-\ln w_i + \ln w_j - \lambda \ln(u_{ij}/m_{ij}) + \eta_{ij} \geq -\ln u_{ij}, i = 1, \dots, n-1; j = i+1, \dots, n. \quad (7)$$

To reach the appropriate solution, the values of the deviation variables should be as small as possible. Therefore, the LFPP-based nonlinear priority model for the FAHP weight derivation is proposed and given by (8) and (9):

$$\text{Minimize } J = (1 - \lambda)^2 + M \cdot \sum_{i=1}^{n-1} \sum_{j=i+1}^n (\delta_{ij}^{*2} + \eta_{ij}^{*2}) \quad (8)$$

$$\begin{aligned} & \text{Subject to} \\ & \begin{cases} x_i - x_j - \lambda \ln\left(\frac{m_{ij}}{l_{ij}}\right) + \delta_{ij} \geq \ln l_{ij}, i = 1, \dots, n-1; j = i+1, \dots, n \\ -x_i + x_j - \lambda \ln\left(\frac{u_{ij}}{m_{ij}}\right) + \eta_{ij} \geq -\ln u_{ij}, i = 1, \dots, n-1; j = i+1, \dots, n \\ \lambda, x_i \geq 0, i = 1, \dots, n \\ \delta_{ij}, \eta_{ij} \geq 0, i = 1, \dots, n-1; j = i+1, \dots, n \end{cases} \end{aligned} \quad (9)$$

Let x_i^* , $i=1, \dots, n$, be the optimal solution to the model proposed (8, 9). The normalized crisp priorities for fuzzy pairwise comparison matrix $\tilde{A} = (\tilde{a}_{ij})_{n \times n}$ can then be obtained as:

$$w_i^* = \frac{e^{x_i^*}}{\sum_{j=1}^n e^{x_j^*}}, i = 1, \dots, n \quad (10)$$

According to Wang and Chin (2011) the LFPP method produces the unique normalized optimal priority vector for any fuzzy pairwise comparison matrix. Therefore, the value of δ^* ($\delta^* = \sum_{i=1}^{n-1} \sum_{j=i+1}^n (\delta_{ij}^{*2} + \eta_{ij}^{*2}) = 0$) can be treated as an inconsistency measure for fuzzy pairwise comparison matrices. The greater value of δ^* , indicates the stronger inconsistency among the fuzzy judgments. It is also preferable to achieve a positive optimal value for λ . Strong inconsistency among the fuzzy judgements can be detected if optimal value λ^* is equal to 0, and the value of δ^* is not equal to 0. However, to check the consistency of responses and matrices created, the most commonly used Consistency Ratio (CR) is calculated.

3.6. Concordance, correlation and compatibility measures

As for the comparison of the results gathered from different stakeholders that can measure the degree of consensus, we employ various indicators, to indicate how diverse are the scorings of respondents. For concordance measure the modified form of the Kendall W calculation (Kendall, 1938; Duleba and Szadoczek, 2022), which provides an overall measure of agreement in ranking is used. In our case, the appropriate procedure is to supplement the rankings provided by the different

stakeholders, and examine the strength of concordance. Kendall W has the common $[0, 1]$ range, where higher values indicate the stronger correlation in ranking. The Kendall W is a non-parametric statistic and it can be calculated as it is explained below.

Let r_{ik} be the rank given to item i by stakeholder k , and R_i (given by (11)) be the aggregated ranking of element i :

$$R_i = \sum_{k=1}^m r_{ik}, i = 1, 2, \dots, n \quad (11)$$

The mean of the aggregated ranking by R can be denoted as (12):

$$R = \frac{m(n+1)}{2} \quad (12)$$

The sum of squares deviation statistic of the aggregated rankings defined by (13):

$$S = \sum_{i=1}^n (R_i - R)^2 \quad (13)$$

The formula for the Kendall coefficient of concordance calculation is given by (14):

$$W = \frac{12S}{m^2(n^3 - n)} \quad (14)$$

When ties occur, W has to be modified by the correction factor (15):

$$T_k = \sum_{i=1}^{g_k} (t_i^3 - t_i) \quad (15)$$

where g_k is the number of groups of ties for stakeholder k and t_i is the number of tied ranks in the i -th group of tied ranks. In this case the corrected Kendall coefficient of concordance can be calculated as given by (16):

$$W = \frac{12 \sum_{i=1}^n (R_i)^2 (n+1)^2}{m^2(n^3 - n) - m \sum_{k=1}^m T_k} \quad (16)$$

We also use the well-known Spearman's rank correlation coefficient (ρ) that is applicable for two vectors (17):

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n^3 - n} \quad (17)$$

where d_i is the difference of the rank of element i for the examined vectors.

When there are ties in the ranks the equation is modified as given by (18):

$$\rho = \frac{\text{cov}(r_1 - r_2)}{\sigma_{r1} \sigma_{r2}} \quad (18)$$

Where r_1 and r_2 denote the rankings defined by the first and second examined vectors, $\text{cov}(r_1 - r_2)$ is the covariance of two variables, while σ is the standard deviation of a variable.

This indicator can take values from the range $[-1, 1]$, thus it can also show the complete disagreement between two preference vectors, compared to the Kendall W measurement.

As for the cardinal indicators, Garuti's compatibility index (G index) (Garuti, 2020) is used. This G index is also applicable for two preference vectors. It considers not only the similarity of the orders but also the weight scores themselves and it is defined by (19):

$$G = \frac{1}{2} \sum_{i=1}^n \frac{\min(w_i^{(1)}, w_i^{(2)})}{\max(w_i^{(1)}, w_i^{(2)})} (w_i^{(1)} + w_i^{(2)}) \quad (19)$$

where $w_i^{(1)}$ and $w_i^{(2)}$ denote the i -th element of the first and second examined vectors. The higher the G index, the more compatible the two respective vectors are.

4. Data and FAHP results

Since air traffic has already been banned on some routes, airlines should consider restructuring their fleet with the introduction of eVTOLs, which would contribute to increased mobility and connectivity, and green aviation, as well. Moreover, eVTOLs could prevent losing market presence, and preserve a certain percentage of air travel demand. However, the capacity of these electric aircraft is smaller compared to aircraft flown the routes before banning. Therefore, ultra-short-haul routes up to 400 km distance and with smaller demand, appear to be proper for an example which will help illustrate the possibilities of our approach.

The two banned routes, Vienna-Graz and Paris (Orly)-Nantes, are taken as examples. These routes were chosen because they are well-known cases of banned ultra-short-haul flights in Europe and have been chosen as representative of the types of connections that could be replaced by more sustainable alternatives such as high-speed rail or eVTOL aircraft. While these routes illustrate the model's application, they do not reflect all possible route types or regional contexts within the European short-haul market. Additionally, these two examples highlight the potential for future multimodal solutions that combine rail and air transport (Buire et al., 2025). The analysis demonstrates how eVTOLs could complement high-speed rail by bridging gaps where rail alone may not fully meet demand or connectivity needs, supporting multimodal networks.

The data on these routes refers to the period from July 19 to July 25 in 2019. The route Vienna-Graz is a 188 km long route, operated by two types of aircraft Bombardier Dash 8 Q400 (DHD8) with a capacity of 76 seats and Embraer 195 (E195) with a configuration of 120 seats. The route Paris (Orly)-Nantes is a longer route compared to the previous one, but still a short-haul route (397 km) operated also by two aircraft types ATR-72-600 (AT76) and Embraer 145 (E145), with the capacities of 70 and 50 seats. The data on these two routes are used to select the most suitable eVTOL from the selected set of six eVTOLs, and to estimate the number of flights and percentage of air travel demand which could be covered by selected eVTOL aircraft for different scenarios.

At the first level of the hierarchy, the importance of each selected criterion should be determined by applying the FAHP and LFPP. The obtained results showed that the highest priority is given to *Passenger capacity* criteria with the 30.50 %, which is followed by *MTOM* with an importance of 25.30 %, *Maximum payload* with 24.51 % and *Range* with 19.69 % of importance (calculated based on fuzzy pairwise comparison matrix given in Table 2).

Fig. 4 illustrates the decomposition of the problem into the hierarchical structure, where six eVTOLs should be ranked based on the selected four criteria to reach the most suitable solution. To apply the proposed methodology, data on each aircraft through all criteria are collected, Table 3.

Table 3

Collected data on selected eVTOL aircraft.

Seat capacity	Seat capacity	MTOM (kg)	Payload capacity (kg)	Range (km)
Joby S4	4	2404	453	161
Lilium Jet	6	3175	500	250
Terafugia TF-2 Lift + Push	4	4080	500	400
Alice	9	7485	1179	402
Jaunt	4	1814	362	182
Embraer Flexcraft	9	3239	1000	926

To apply FAHP, it is necessary to define fuzzy pairwise comparison matrices of eVTOLs with respect to selected criteria. Based on the data from Table 3 and according to the authors' expert opinions, these fuzzy pairwise comparison matrices are given in Tables 4–7. Table 4 is created based on the assumption that a higher *Seat capacity* of an eVTOL aircraft corresponds to greater preference and, consequently, higher relative importance compared to other eVTOLs. The consistency of the pairwise comparisons has been verified, yielding a $CR=0.022$. Since this value is well below the commonly accepted threshold of 0.10, the judgments can be considered consistent and reliable for further analysis.

Based on the fuzzy pairwise comparison matrix (Table 4), local priorities with respect to c_1 (*Seat capacity*) are calculated and given in Table 8. Alice and Embraer Flexcraft are the best solutions with respect to this criterion since they have the largest capacity, and the model aims

Table 4

Fuzzy pairwise comparison matrix of eVTOLs with respect to the *Seat capacity*.

Seat capacity	Joby S4	Lilium Jet	Terafugia TF-2 Lift + Push	Alice	Jaunt	Embraer Flexcraft
Joby S4	(1, 1, 1)	(1/2, 2/3, 1)	(1/2, 1, 3/2)	(2/5, 1/2, 2/3)	(1/2, 1, 3/2)	(2/5, 1/2, 2/3)
Lilium Jet	(1, 3/2, 2, 2)	(1, 1, 1)	(1, 3/2, 2)	(1/2, 2/3, 1)	(1, 3/2, 2, 2)	(1/2, 2/3, 1)
Terafugia TF-2 Lift + Push	(2/3, 1, 2)	(1/2, 2/3, 1)	(1, 1, 1)	(2/5, 1/2, 2/3)	(1/2, 1, 3/2)	(2/5, 1/2, 2/3)
Alice	(3/2, 2, 5/2)	(1, 3/2, 2)	(3/2, 2, 5/2)	(1, 1, 1)	(2, 5/2, 3)	(1/2, 1, 3/2)
Jaunt	(2/3, 1, 2)	(1/2, 2/3, 1)	(2/3, 1, 2)	(1/3, 2/5, 1/2)	(1, 1, 1)	(2/5, 1/2, 2/3)
Embraer Flexcraft	(3/2, 2, 5/2)	(1, 3/2, 2)	(3/2, 2, 5/2)	(2/3, 1, 2)	(3/2, 2, 5/2)	(1, 1, 1)

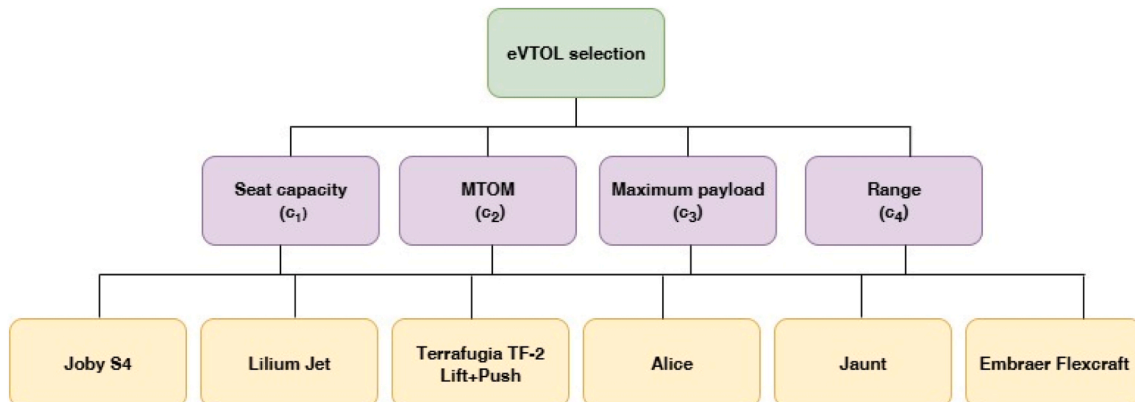


Fig. 4. The hierarchical structure of eVTOL selection problem.

Table 5Fuzzy pairwise comparison matrix of eVTOLs with respect to the *MTOM*.

MTOM	Joby S4	Lilium Jet	Terafugia TF-2 Lift + Push	Alice	Jaunt	Embraer Flexcraft
Joby S4	(1, 1, 1)	(1, 3/2, 2)	(3/2, 2, 5/2)	(2, 5/2, 3)	(1/2, 2/3, 1)	(1, 3/2, 2)
Lilium Jet	(1/2, 2/3, 1)	(1, 1, 1)	(1, 3/2, 2)	(3/2, 2, 5/2)	(2/5, 1/2, 2/3)	(1/2, 1, 3/2)
Terafugia TF-2 Lift + Push	(2/5, 1/2, 2/3)	(1/2, 2/3, 1)	(1, 1, 1)	(1, 3/2, 2)	(1/3, 2/5, 1/2)	(1/2, 2/3, 1)
Alice	(1/3, 2/5, 1/2)	(2/5, 1/2, 2/3)	(1/2, 2/3, 1)	(1, 1, 1)	(2/7, 1/3, 2/5)	(2/5, 1/2, 2/3)
Jaunt	(1, 3/2, 2)	(3/2, 2, 5/2)	(2, 5/2, 3)	(5/2, 3, 7/2)	(1, 1, 1)	(3/2, 2, 5/2)
Embraer Flexcraft	(1/2, 2/3, 1)	(2/3, 1, 2)	(1, 3/2, 2)	(3/2, 2, 5/2)	(2/5, 1/2, 2/3)	(1, 1, 1)

Table 6Fuzzy pairwise comparison matrix of eVTOLs with respect to the *Maximum payload*.

Maximum payload	Joby S4	Lilium Jet	Terafugia TF-2 Lift + Push	Alice	Jaunt	Embraer Flexcraft
Joby S4	(1, 1, 1)	(1, 1, 1)	(1, 1, 1)	(2/5, 1/2, 2/3)	(1, 3/2, 2)	(2/5, 1/2, 2/3)
Lilium Jet	(1, 1, 1)	(1, 1, 1)	(1, 1, 1)	(2/5, 1/2, 2/3)	(1, 3/2, 2)	(2/5, 1/2, 2/3)
Terafugia TF-2 Lift + Push	(1, 1, 1)	(1, 1, 1)	(1, 1, 1)	(2/5, 1/2, 2/3)	(1, 3/2, 2)	(2/5, 1/2, 2/3)
Alice	(3/2, 2, 5/2)	(3/2, 2, 5/2)	(3/2, 2, 5/2)	(1, 1, 1)	(2, 5/2, 3)	(1, 1, 1)
Jaunt	(1/2, 2/3, 1)	(1/2, 2/3, 1)	(1/2, 2/3, 1)	(1/3, 2/5, 1/2)	(1, 1, 1)	(1/3, 2/5, 1/2)
Embraer Flexcraft	(3/2, 2, 5/2)	(3/2, 2, 5/2)	(3/2, 2, 5/2)	(1, 1, 1)	(2, 5/2, 3)	(1, 1, 1)

Table 7Fuzzy pairwise comparison matrix of eVTOLs with respect to the *Range*.

Range	Joby S4	Lilium Jet	Terafugia TF-2 Lift + Push	Alice	Jaunt	Embraer Flexcraft
Joby S4	(1, 1, 1)	(1/2, 2/3, 1)	(1/3, 2/5, 1/2)	(1/3, 2/5, 1/2)	(1/2, 1, 3/2)	(2/7, 1/3, 2/5)
Lilium Jet	(1, 3/2, 2)	(1, 1, 1)	(2/5, 1/2, 2/3)	(2/5, 1/2, 2/3)	(1, 3/2, 2)	(1/3, 2/5, 1/2)
Terafugia TF-2 Lift + Push	(2, 5/2, 3)	(3/2, 2, 5/2)	(1, 1, 1)	(2/3, 1, 2)	(3/2, 2, 5/2)	(1/2, 2/3, 1)
Alice	(2, 5/2, 3)	(3/2, 2, 5/2)	(1/2, 1, 3/2)	(1, 1, 1)	(3/2, 2, 5/2)	(1/2, 2/3, 1)
Jaunt	(2/3, 1, 2)	(1/2, 2/3, 1)	(2/5, 1/2, 2/3)	(2/5, 1/2, 2/3)	(1, 1, 1)	(2/7, 1/3, 2/5)
Embraer Flexcraft	(5/2, 3, 7/2)	(2, 5/2, 3)	(1, 3/2, 2)	(1, 3/2, 2)	(5/2, 3, 7/2)	(1, 1, 1)

Table 8

Aggregation of the local priorities.

	c_1 (0.3050)	c_2 (0.2530)	c_3 (0.2451)	c_4 (0.1969)	Global LFPP priorities
<i>Local priorities of selected alternatives with respect to the selected criteria</i>					
Joby S4	0.1176	0.2159	0.1361	0.0900	0.1416
Lilium Jet	0.1765	0.1440	0.1361	0.1162	0.1465
Terafugia TF-2 Lift + Push	0.1176	0.1148	0.1361	0.2087	0.1394
Alice	0.2353	0.0934	0.2108	0.2250	0.1914
Jaunt	0.1176	0.2879	0.1089	0.0900	0.1531
Embraer Flexcraft	0.2353	0.1440	0.2721	0.2700	0.2281

to maximize it (local priority is equal to 0.2353).

In the fuzzy pairwise comparison matrix of eVTOLs with respect to *MTOM*, aircraft with a lower *MTOM* are considered more preferable. This is because *MTOM* serves as a primary basis for calculating airport and air navigation service fees. The fuzzy pairwise comparison matrix was found to be consistent, with a $CR = 0.016$, which is well within the acceptable threshold ($CR < 0.10$), indicating reliable judgments. Jaunt has the largest local priority (0.2879) with respect to *MTOM* (Table 8, calculated based on Table 5), and therefore it is the best solution regarding c_2 .

Since *Maximum payload* reflects the available capacity for passengers and cargo, it can be interpreted as an indicator of service level. Accordingly, a higher *Maximum payload* implies greater relative importance when compared to other eVTOLs (Table 6). After verifying the consistency of the fuzzy pairwise comparison matrix ($CR = 0.008$), the local priorities were calculated with respect to c_3 (*Maximum payload*) and are presented in Table 8. Embraer Flexcraft is, again, the best solution with respect to this criterion (0.2721).

Table 7 is based on the premise that *Range* (measured in km) represents an aircraft's ability to operate over a certain distance without the need for battery recharging. Consequently, a greater *Range* indicates higher operational flexibility, making an eVTOL aircraft more preferable relative to others. The consistency of the fuzzy pairwise comparison matrix was confirmed, ($CR = 0.018$), indicating acceptable consistency of judgments. In this context, the Embraer Flexcraft emerges as the most favourable option in terms of *Range*, achieving the highest score (0.2700).

Besides all local priorities, Table 8 provides global crisp priorities, as well. It can be seen that Embraer Flexcraft is the most suitable eVTOL for ultra-short-haul routes. The final rank of alternatives is: Embraer Flexcraft (22.81 %), Alice (19.14 %), Jaunt (15.31 %), Lilium Jet (14.65 %), Joby S4 (14.16 %), and Terafugia TF-2 Lift + Push (13.94 %).

5. Analysis and discussion

This section presents a series of analyses followed by a corresponding discussion. First, the sensitivity of the eVTOL aircraft rankings and the consistency ratios with respect to variations in judgments within the fuzzy pairwise comparison matrix at the first hierarchical level is examined. Next, stakeholder groups are formed from the survey respondents to evaluate potential differences in their prioritization of criteria and rankings of eVTOL alternatives. Finally, two illustrative examples of ultra-short-haul routes are provided to demonstrate the potential passenger demand that could be attracted under such services.

5.1. Sensitivity analysis of FAHP solutions

After reaching the solution (by applying FAHP), an electric aircraft that fits the defined requirements best, the sensitivity of the solution (rank of eVTOL aircraft) and the consistency ratio concerning different judgement in a fuzzy pairwise comparison matrix for the first level are analyzed. By varying one element in the fuzzy pairwise matrix according

to the triangular fuzzy scale, while keeping the other elements constant, the consistency ratio is calculated as well as the final priority vector that defines the rank of alternatives. The analysis encompasses 43 experiments (including starting solution) that achieve the appropriate values of the consistency ($CR < 0.1$ in all experiments). It should be noted that the initial analysis included 37 experiments that met the required level of consistency, as assessed using the method proposed by Wang and Chin (2011). However, after computing the CR, six additional matrices were found to be consistent. This outcome suggests that, in this particular case (though the authors do not claim this as a general rule), the consistency criterion defined by Wang and Chin (2011) appears to be more stringent than the traditional CR-based approach.

The sensitivity analysis (Fig. 5) shows that the final rank of six eVTOLs is almost insensitive to this kind of change. Namely, the most suitable is always Embraer Flexcraft, and the second-ranked is Alice. The third one is most commonly Jaunt (third-ranked in 36 experiments). Lilium Jet appears as the third solution in six experiments, and Terafugia TF-2 Lift + Push was third-ranked eVTOL aircraft once. The rank four most often belongs to Lilium Jet (in 27 experiments), while rank five in 22 experiments belongs to Joby S. The rank six is shared between Terafugia TF-2 Lift + Push (27 experiments) and Joby S (16 experiments).

Assuming that each experiment in the sensitivity analysis represents the judgment of a different evaluator assessing the selected criteria, Kendall's coefficient of concordance (W) can be applied to measure the level of agreement among evaluators when ties exist (see equation (16) in Section 3.6). The resulting value is $W = 0.577$, with a p -value of 4.858×10^{-16} , which is below the 0.05 significance threshold. This indicates that while the result is statistically significant, the degree of agreement among evaluators is moderate. However, when the same measure is applied to the rankings of eVTOL alternatives, a substantially different result emerges. In this case, the Kendall's $W = 0.902$ (see equation (14) in Section 3.6), with a p -value of 5.48007×10^{-40} , also below 0.05. This high level of concordance allows us to reject the null hypothesis of no agreement, indicating strong consensus among evaluators regarding the ranking of eVTOL alternatives.

5.2. Concordance, correlation and compatibility measures of stakeholders' opinions

Following the sensitivity analysis, which confirmed the stability of the FAHP solution and its robustness to changes in the relative importance of criteria, further analysis was conducted to examine how different stakeholder groups valued the various criteria. In addition, measures of concordance, correlation, and compatibility were assessed to better understand the alignment of preferences. As mentioned in Section 3.3, the experts participating in the survey were categorized into seven stakeholder groups: manufacturers, private companies, airlines, academia, regulatory bodies, ANSPs, and drone pilots. Since the groups representing regulatory bodies, ANSPs, and drone pilots included only one respondent each, the AHP was applied to derive the final ranking of eVTOL alternatives for these groups. For the remaining groups, which included multiple respondents, the FAHP method was employed to account for uncertainty and variation in judgments. The resulting priority vectors and CR for all stakeholder groups and the solution derived by FAHP, are summarized in Table 9.

The value of Kendall's coefficient of concordance among the seven stakeholder groups is $W = 0.703$ indicating that stakeholders shared substantially similar judgments. The corresponding p -value = 0.002

Table 9
Criteria local priorities and CR values across stakeholder groups and the FAHP solution.

	Seat capacity	MTOM	Payload capacity	Range	CR
FAHP solution	0.3050	0.253	0.2451	0.1969	0.045
Manufacturers	0.2933	0.2525	0.2271	0.2271	0.060
Private companies	0.3031	0.2474	0.2474	0.2020	0.060
Airlines	0.1666	0.1667	0.2503	0.4165	0.026
Academia	0.2126	0.1888	0.2515	0.3470	0.039
Regulatory body	0.4400	0.3740	0.1400	0.0460	0.058
ANSP	0.3825	0.3825	0.1750	0.0600	0.057
Drone pilot	0.7400	0.0480	0.1060	0.1060	0.057

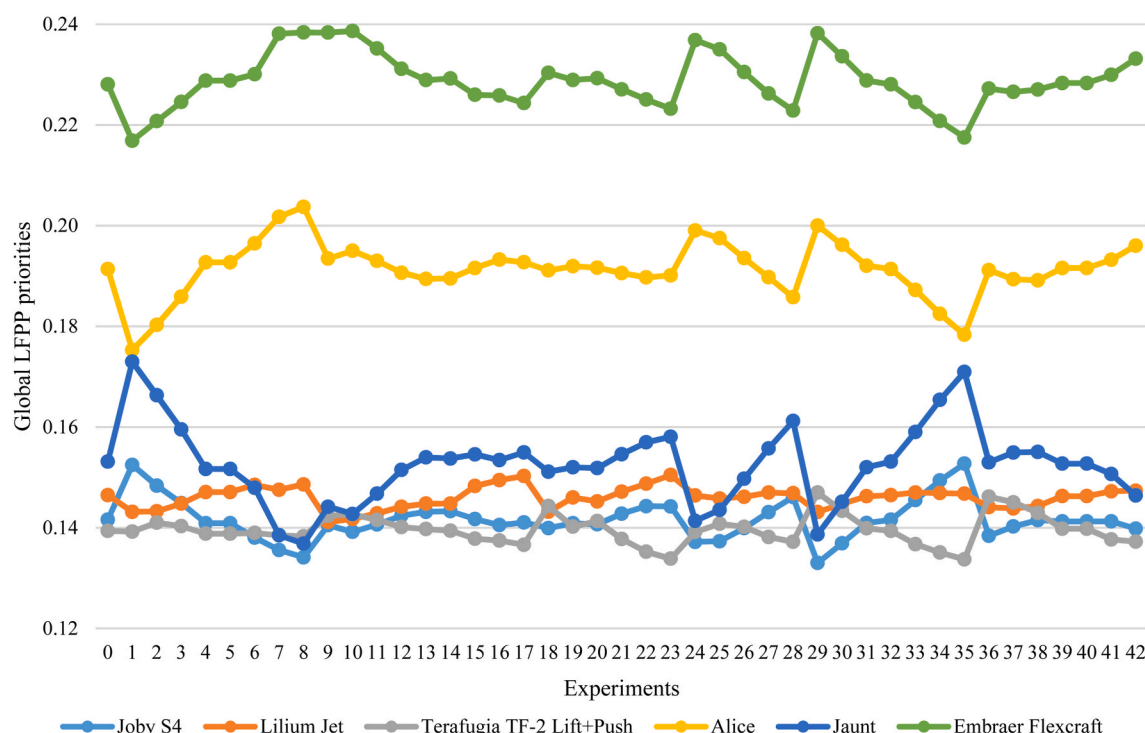


Fig. 5. The sensitivity of the solution – rank of eVTOLs.

confirms that this agreement is statistically significant. When the FAHP-derived solution is included as the eighth member in the set of stakeholders, then the value of W is slightly decreased to $W=0.69$, with a p -value = 0.001 which still indicates a statistically significant and strong level of concordance.

To further assess the alignment between individual stakeholder preferences and the FAHP-derived solution, we employ the well-known Spearman's rank correlation coefficient (Spearman's ρ , calculated according equation (17) or (18), Section 3.6). This non-parametric measure, suitable for comparing rankings between two evaluators, ranges from -1 (complete disagreement) to $+1$ (complete agreement), thus allowing us to evaluate the degree of correspondence between each stakeholder group's ranking and the FAHP-based result. In addition, Garuti's compatibility index (G -index) (Garuti, 2020) is calculated to complement this analysis (see equation (19) in Section 3.6). Unlike Spearman's ρ , the G -index accounts not only for the order of preferences but also for the relative weight values assigned by each stakeholder group compared to the FAHP solution. Table 10 presents the values of Spearman's rank correlation coefficient (ρ) and Garuti's compatibility index (G -index) for each stakeholder group, comparing their criteria rankings and weights to the FAHP-derived solution.

From the table, it is evident that *Manufacturers*, *Private companies*, and *ANSPs* exhibit high Spearman's ρ values, indicating a very high correlation with the FAHP solution, with *Regulatory bodies* showing perfect agreement ($\rho=1$). In contrast, *Airlines* and *Academia* display strong negative correlations (-0.947 and -0.800 , respectively), indicating substantial disagreement with the FAHP-derived rankings. *Drone pilots*, with $\rho = 0.513$, show a moderate level of correlation.

Regarding Garuti's G -index (Garuti, 2020), *Manufacturers* and *Private companies* demonstrate very high compatibility with the FAHP solution ($G > 0.9$), while *Airlines*, *Academia*, and *ANSPs* fall within a low compatibility range ($0.65 < G < 0.75$). The *Regulatory bodies* displays low compatibility ($0.60 < G < 0.65$), despite having perfect rank correlation—highlighting a divergence in weight magnitudes rather than order. Finally, the *Drone pilots* group, with $G < 0.6$, reflects a random level of compatibility, suggesting minimal alignment with the FAHP-derived weights.

Table 11 presents global priority vectors for each group of stakeholders, and FAHP-derived global priority vectors. Now, it is possible to calculate Kendall's coefficient of concordance, Spearman's rank correlation coefficient (ρ) and Garuti's compatibility index (G -index) for each stakeholder group, comparing their final rankings of eVTOLs and weights to the FAHP-derived solution.

The value of Kendall's coefficient of concordance among the seven stakeholder groups is $W = 0.753$ indicating that stakeholders shared substantially similar judgments. The corresponding p -value = 7.64×10^{-5} confirms that this agreement is statistically significant. When the FAHP-derived solution is included as the eighth member in the set of stakeholders, then the value of W is slightly increased to $W=0.770$, with a p -value = 1.03×10^{-5} which still indicates a statistically significant and strong level of concordance.

Table 12 shows the values of ρ and G -index for stakeholders evaluations of alternatives. It can be observed that *Manufacturers*, *Regulatory bodies*, and *ANSPs* recorded high Spearman's ρ values, indicating a very high correlation with the FAHP solution, with *Private companies* showing perfect agreement

($\rho=1$). In contrast, *Airlines* and *Academia* still display the lowest values of ρ , indicating moderate correlation, while *Drone pilots*, show a high level of correlation.

Regarding Garuti's G -index (Garuti, 2020), all stakeholders except the *Drone pilots* demonstrate very high compatibility with the FAHP solution ($G > 0.9$). The *Drone pilots* group, with $G < 0.885$ ($0.85 < G < 0.9$), reflects a high compatibility with the FAHP-derived weights.

This analysis further reinforces the findings of the sensitivity analysis: the FAHP-derived solution, which identifies the Embraer Flexcraft as the most suitable option for European ultra-short-haul routes, is consistently confirmed across all stakeholder groups as the most appropriate alternative. Similar to the principle in commercial aviation, capacity appears to be one of the most important factors when choosing the adequate aircraft. Thus, our analysis approved this well-established hypothesis by consistently favouring the alternative (i.e., vehicle) which owns the best characteristics with respect to selected criteria. This suggests that even small changes in less influential criteria will not significantly affect the overall ranking which provides decision-makers more confidence in selecting the preferred alternative.

5.3. Potential passenger demand on routes analysis

Recent studies on urban air mobility showed that eVTOLs can accommodate up to 4 % of intra-city travels (Straubinger et al., 2020). Bearing in mind that the two selected routes are inter-city routes, with previously solid demand for air services, it is expected that they could attract more than 4 % of travellers who shifted to rail following the bans. Table 13 presents realised traffic for the considered period on the Vienna-Graz route (data are taken from Flightradar24). The number of passengers ($No\ pax$) was estimated based on aircraft capacity (DH8D and E195 types), daily frequencies, and an assumed 80 % load factor. This assumption reflects the average domestic load factor for Austrian Airlines in 2019 (Austrian Airlines, 2019), which operated all flights on the route (Table 13).

Using these data, frequencies for potential eVTOL operations were estimated for different demand scenarios. In the baseline case, where service frequency remains equal to that prior to the ban, the Embraer Flexcraft would capture approximately 12 % of passengers on the route. To maintain weekly frequency, the airline would therefore need to re-attract about 12 % of those who shifted to rail. If the response to the new service is weaker, e.g., only 10 % of former passengers return, then total frequencies would be lower, and operations could be covered by a single Embraer Flexcraft eVTOL (Table 13). Conversely, in a more optimistic scenario where 20 % of passengers are re-attracted, frequencies would exceed those before the ban, and a single aircraft would not be sufficient. In this case, the fleet would need to be expanded to at least two eVTOLs. Depending on the airline's policy, and its business goals, the decision related to frequency should be made. Because the frequency is a very important factor from passengers' perspective (when choosing an airline), an airline should keep the same or increase it. However, the significant increase in frequency could influence ATM and its complexity, especially due to regulations related to eVTOLs that are not in place yet.

Another analysis is based on different average load factors (LF). Namely, the estimated load factor of 80 % is changed in the interval of 5 % decrease and increase (Fig. 6). The expected weekly number of passengers varies from 259 to 284 in the case of 10 % of air travel demand shifted to eVTOLs, and from 2592 to 2835 in the case of 100 % shifted passengers.

Regarding weekly frequencies (Fig. 7), minimal weekly frequency (in the case of 10 % attracted air travel demand) is in the range 29–32 for considered LF s, while maximal weekly frequency is in the range 288–315. If an airline intends to keep weekly frequency constant and equal to the weekly frequency before the ban (40 flights), it is needed to attract around 12 % of shifted passengers (for $LF = 85\%$) or around 13 % (for $LF = 75\%$).

Table 10
Spearman's ρ and G index for criteria.

	Spearman ρ	G index
Manufacturers	0.947	0.942
Private companies	0.923	0.985
Airlines	-0.947	0.655
Academia	-0.800	0.742
Regulatory bodies	1	0.609
ANSPs	0.884	0.673
Drone pilots	0.513	0.401

Table 11
eVTOLs global priority weights across stakeholder groups and the FAHP solution.

	Joby S4	Lilium Jet	Terafugia TF-2 Lift + PushAlice	Alice	Jaunt	Embraer Flexcraft
FAHP solution	0.1416	0.1465	0.1394	0.1914	0.1531	0.2281
Manufacturers	0.1404	0.1454	0.1418	0.1916	0.1523	0.2285
Private companies	0.1409	0.1463	0.1399	0.1920	0.1520	0.2289
Airlines	0.1271	0.1359	0.1597	0.2012	0.1323	0.2437
Academia	0.1312	0.1393	0.1533	0.1988	0.1380	0.2394
Regulatory bodies	0.1557	0.1559	0.1233	0.1783	0.1788	0.2079
ANSPs	0.1568	0.1534	0.1252	0.1761	0.1796	0.2089
Drone pilots	0.1214	0.1643	0.1291	0.2248	0.1219	0.2385

Table 12
Spearman's ρ and G index for eVTOLs.

	Spearman ρ	G index
Manufacturers	0.943	0.994
Private companies	1	0.996
Airlines	0.600	0.913
Academia	0.600	0.937
Regulatory bodies	0.943	0.907
ANSPs	0.886	0.908
Drone pilots	0.714	0.885

Table 13
Route Vienna-Graz operated by Embraer Flexcraft (9 seats).

Date	Realised traffic			Estimated air travel demand and frequencies					
	DH8D frequency (76 seats)	E195 frequency (120 seats)	No pax (LF = 80 %)	10 % of pax	Embraer Flexcraft frequency (10 % pax)	20 % of pax	Embraer Flexcraft frequency (20 % pax)	12 % of pax	Frequency equals to total frequency before ban
July 19, 2019.	5	1	400	40	5	80	9	48	6
July 20, 2019.	3	1	278	28	4	56	7	34	4
July 21, 2019.	3	1	278	28	4	56	7	34	4
July 22, 2019.	6	0	365	37	5	73	9	46	5
July 23, 2019.	5	3	592	60	7	119	14	71	9
July 24, 2019.	5	1	400	40	5	80	9	48	6
July 25, 2019.	5	1	400	40	5	80	9	48	6

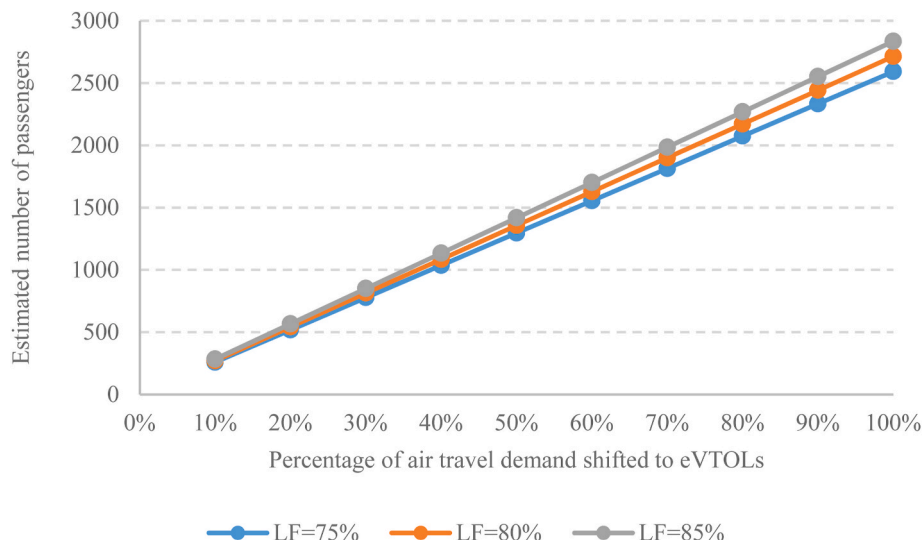


Fig. 6. Estimated number of passengers depending on LF and percentage of shifted air travel demand – Vienna-Graz.

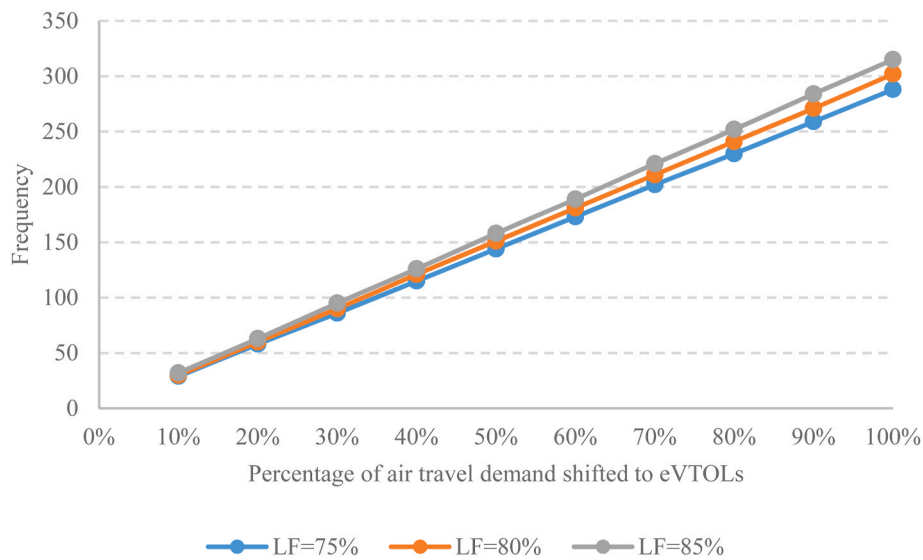


Fig. 7. Estimated frequencies depending on LF and percentage of shifted air travel demand – Vienna-Graz.

Table 14

Route Paris (Orly)-Nantes operated by Embraer Flexcraft (9 seats).

Date	Realised traffic			Estimated air travel demand and frequencies					
	AT76 frequency (70 seats)	E145 frequency (50 seats)	No pax (LF = 85 %)	10 % of pax	Embraer Flexcraft frequency (9 seats)	20 % of pax	Embraer Flexcraft frequency (9 seats)	15 % of pax	Frequency before ban (Embraer Flexcraft - 9 seats)
July 19, 2019.	4	2	323	33	4	65	8	49	6
July 20, 2019.	6	0	357	36	4	72	8	54	6
July 21, 2019.	6	0	357	36	4	72	8	54	6
July 22, 2019.	4	2	323	33	4	65	8	49	6
July 23, 2019.	4	2	323	33	4	65	8	49	6
July 24, 2019.	4	2	323	33	4	65	8	49	6
July 25, 2019.	4	2	323	33	4	65	8	49	6

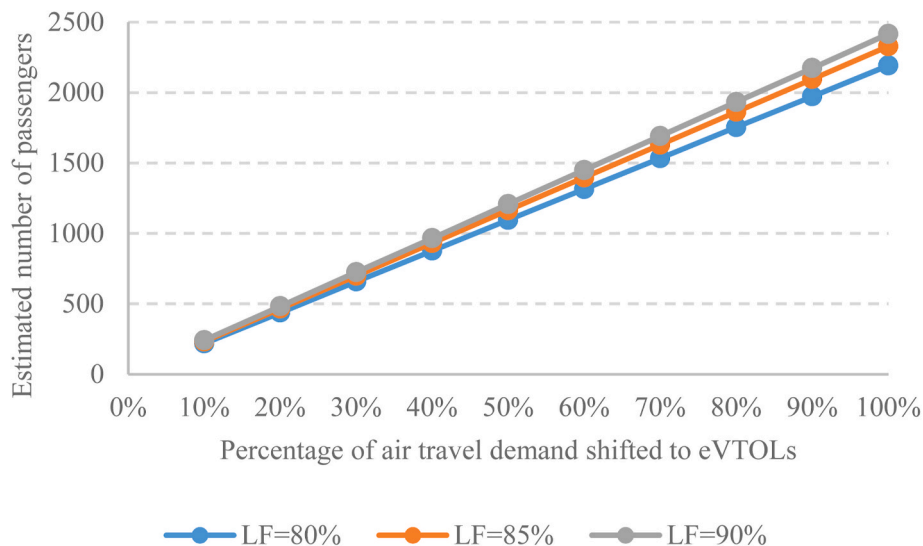


Fig. 8. Estimated number of passengers depending on LF and percentage of shifted air travel demand – Paris (Orly)-Nantes.

Estimated frequencies depending on LF and percentage of shifted air travel demand for the Paris (Orly)-Nantes route are shown in Fig. 9. It can be seen that the minimal weekly frequency (in the case of 10 % attracted air travel demand) is in the range of 24–28 for considered LF s, while the maximal weekly frequency is in the range of 244–268. If an airline intends to keep weekly frequency constant and equal to the weekly frequency before the ban (42 flights), it is needed to attract around 14 % of shifted passengers (for $LF = 90$ %) or around 16 % (for $LF = 80$ %).

Since the second ranked, Alice eVTOL, has the same capacity as Embraer Flexcraft, the analysis shown could be applied to this eVTOL aircraft as well. The examples of these two routes showed that airlines could keep their presence on banned routes by introducing green fleets, eVTOLs that do not pollute the air, but increase air mobility. However, it is on an airline to select the strategy and to make its decisions in accordance with that.

Many travelers still do not view rail as attractive, since airlines better meet expectations for price, convenience, and comfort. The proposed eVTOL capacity is lower than pre-ban airline capacity, but operators should be able to fill it. Travel habits, attitudes, and demographic factors (e.g., income, employment) strongly influence mode choice, with higher-income travelers often preferring air for its convenience. Passengers are likely to shift to new air services when benefits such as speed, comfort, accessibility, and frequency outweigh costs. The main uncertainty lies in whether operators can offer competitive prices, as eVTOL operating costs remain unclear. A precise demand forecast would require a comprehensive behavioral study, which is beyond the scope of this research and acknowledged as a limitation.

6. Conclusion

The ban on ultra-short-haul flights and their shift to rail for environmental protection has its advantages, such as significantly less pollution, providing more airport resources to be used for other purposes, and developing other modes of transport. However, this also brings some negative effects. For instance, it may lessen mobility and accessibility by cutting the most efficient and fastest mode of transport, directly affecting the regional economy, since this ban could result in job losses and tourism reduction. Moreover, uncertainty about passenger rights, fragmented operations, infrastructure bottlenecks, a national focus, and customer experience issues, all limit the high-speed rail's potential as a strong alternative.

The aim of this paper is to bridge a gap between the two opposite

approaches and offer a compromise solution. The recent advances in the development of electric and hydrogen-powered propulsive systems enabled the emergence of eVTOL vehicles, which are seen as a sustainable alternative to traditional fuel-powered aircraft. Moreover, eVTOL aircraft operations are foreseen to offer tangible benefits to both inhabitants and the economy.

This paper offers the methodology for the selection of a new type of vehicle with zero emissions. Although the aircraft selection problem is a well-known and investigated problem in the academic literature, the approach presented in this paper provides new insight into the new types of vehicles in the future fleet and pays special attention to the aspect of environmental protection. The main contributions of this paper are threefold: Firstly, it involves experts' judgment in the process of ranking a set of criteria to be used as input in the model. Secondly, it establishes a connection between the concept of UAM and banned air routes as possible solutions for airlines. Finally, it assesses air travel demand and frequency to demonstrate the potential of eVTOLs operating on banned European routes. For this purpose, a number of different eVTOL aircraft are used and tested through the selected criteria to help perceive and explore opportunities of the electrical aircraft that will be needed to support ultra-short-haul operations.

Additionally, the FAHP method has proved to be very useful in solving a variety of choice problems, including aircraft selection problems. However, leveraging the input obtained from experts from various fields, the study ensured a comprehensive understanding of the criteria and challenges associated with the evaluation of eVTOL vehicles, reflecting the collective expertise and addressing the complex, multi-faceted nature of the field.

It is important to underline that the study has certain limitations. Most notably, it lacks operational cost data, which is a crucial factor in aircraft purchasing decisions. Moreover, the analysis assumes that eVTOLs are operationally ready, that the necessary infrastructure (vertiports, charging stations, and airspace management systems) is available, and that the public is willing to adopt this new mode of transport. In practice, eVTOL aircraft deployment remains at an early stage, with limited pilot projects, test flights, and certification processes underway, while vertiport networks are still emerging and face regulatory, technical, and land-use challenges. Public acceptance also remains uncertain, as concerns over noise, safety, privacy, cost, and integration with existing transport systems may influence willingness to use eVTOL services. Therefore, the assumptions made in this study may not hold in practice, potentially affecting the feasibility and scalability of the proposed solutions. Finally, it is important to note that the assessment of air

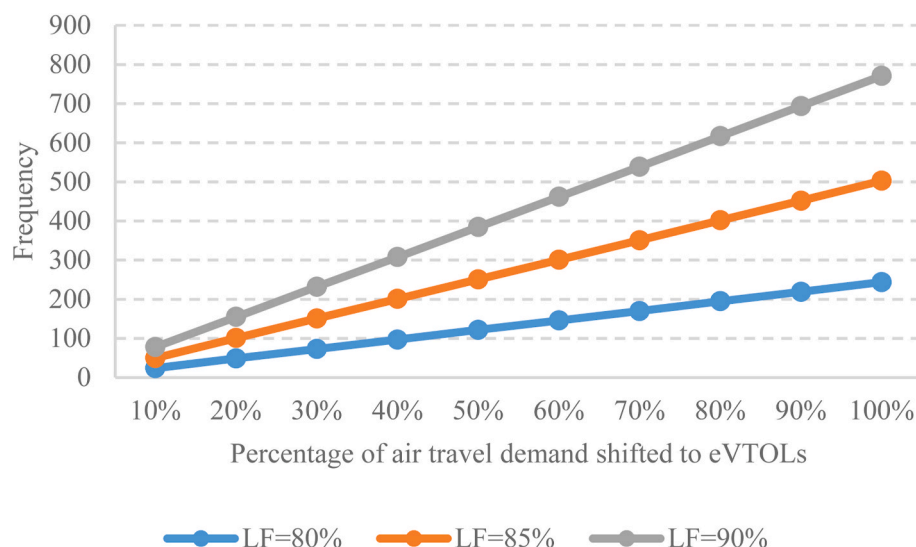


Fig. 9. Estimated frequencies depending on LF and percentage of shifted air travel demand – Paris (Orly)-Nantes.

travel demand is still based on the authors' estimations, as precise data is currently unavailable. A comprehensive demand analysis would require detailed socio-economic, modal-choice, and behavioral data, which are beyond the scope of the current study. Nevertheless, this analysis provides a valuable first insight on the problem, offering an initial exploration of the potential for eVTOL aircraft integration. Demonstrating their opportunities and concept of operations is the first step toward the implementation of eVTOLs.

As more data becomes available, the proposed methodology can be adapted to include additional criteria, operational costs, and refined demand estimations, providing decision-makers with an enriched perspective grounded in real-world exploitation data. While the present analysis and scenarios involve significant uncertainty—given that the UAM sector is still under development, demand projections are preliminary, and no definitive business model has been established—this study highlights the potential of UAM as a fast-growing area of interest for both academia and industry. Proactively analyzing UAM's trajectory is essential to ensure its sustainable growth and fulfil technical, economic, and environmental expectations, even as scenarios may evolve in unpredictable directions.

CRedit authorship contribution statement

Slavica Dožić: Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Jovana Kuljanin:** Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Danica Babić:** Writing – original draft, Methodology, Formal analysis. **Milica Kalić:** Validation, Methodology, Conceptualization. **Jordi Pons-Prats:** Supervision, Resources, Project administration, Data curation.

Acknowledgement

This research was supported by grants PID2023-152053OB-C21, funded by MICIU/AEI /10.13039/501100011033 and by FEDER, UE.

Data availability

Data will be made available on request.

References

- AiRMOUR, 2023. Guidebook for urban air Mobility integration. AiRMOUR Deliverable 6 (4). Available online at: https://www.easa.europa.eu/sites/default/files/dfu/airmour_d6.4_guidebook_for_uam_integration_process_management.pdf. (Accessed 20 August 2025).
- Anger, A., Köhler, J., 2010. Including aviation emissions in the EU ETS: much ado about nothing? A review. *Transp. Policy* 17 (1), 38–46.
- Austrian airlines. <https://www.austrianairlines.ag/en/2019/09/12/traffic-results-for-august-2019-1-5-million-passengers-flew-with-austrian-airlines-in-august/>, 2019 (accessed 7.07.23).
- Barzkar, A., Ghassemi, M., 2020. Electric power systems in more and all electric aircraft: a review. *IEEE Access* 8, 169314–169332.
- Bridgelall, R., Askarzadeh, T., Tolliver, D.D., 2023. Introducing an efficiency index to evaluate eVTOL designs. *Technol. Forecast. Soc. Change* 191, 122539.
- Bruno, G., Esposito, E., Genovese, A., 2015. A model for aircraft evaluation to support strategic decisions. *Expert Syst. Appl.* 42 (13), 5580–5590.
- Buire, C., Dožić, S., Babić, D., Stroumpou, I., Larriba, J.L., Gatein, E., Parajó, R., 2025. Air–Rail connectivity index: a comprehensive Study of multimodal journeys. *Engineering Proceedings* 90 (1), 72. <https://doi.org/10.3390/engproc2025090072>.
- Chopra, I.P., Jain, T., Gupta, S., 2025. Prioritising factors affecting online shopping behaviour: a fuzzy AHP approach. *Int. J. Internet Market Advert.* 22 (2), 187–213.
- Clark, P., 2007. *Buying the Big Jets: Fleet Planning for Airlines*, second ed. Routledge, New York.
- Dawar, V.K., Gupta, S., Bajaj, A., Gupta, M., 2025. Unveiling ESG realities: fuzzy AHP assessment of greenwashing in investor portfolios. *South Asian Journal of Business Studies*.
- Deloitte, 2019. Change is in the air. The elevated future of mobility: what's next on the horizon? <https://www2.deloitte.com/us/en/insights/focus/future-of-mobility/evto1-elevated-future-of-mobility-summary.html> (accessed on 17.03.23).
- Dobruszkes, F., Mattioli, G., Mathieu, L., 2022. Banning super short-haul flights: environmental evidence or political turbulence? *J. Transport Geogr.* 104, 103457.
- Dožić, S., 2019. Multi-criteria decision making methods: application in the aviation industry. *J. Air Transport. Manag.* 79, 101683.
- Dožić, S., Kalić, M., 2014. An AHP approach to aircraft selection process. *Transp. Res. Procedia* 3, 165–174.
- Dožić, S., Kalić, M., 2015. Three-stage airline fleet planning model. *J. Air Transport. Manag.* 46, 30–39.
- Dožić, S., Kalić, M., 2015a. Comparison of two MCDM methodologies in aircraft type selection problem. *Transp. Res. Procedia* 10, 910–919.
- Dožić, S., Lutovac, T., Kalić, M., 2018. Fuzzy AHP approach to passenger aircraft type selection. *J. Air Transport. Manag.* 68, 165–175.
- Duleba, S., Szadoczi, Z., 2022. Comparing aggregation methods in large-scale group AHP: time for the shift to distance-based aggregation. *Expert Syst. Appl.* 196, 116667.
- Electrical VTOL News, 2024. eVTOL Aircraft Directory. <https://evtol.news/aircraft>. (Accessed 20 December 2024).
- Eker, U., Fountas, G., Anastasopoulos, P.C., Still, S.E., 2020. An exploratory investigation of public perceptions towards key benefits and concerns from the future use of flying cars. *Travel Behaviour and Society* 19, 54–66.
- Eker, U., Fountas, G., Anastasopoulos, P.C., 2019. A Preliminary Investigation of Public Expectations and Interest in Flying Cars: a Statistical Analysis of Willingness to Pay and Willingness to Use Flying Cars. Working paper.
- Euractiv, 2023. Airlines to invoke EU freedoms to challenge green restrictions on short-haul flights. <https://www.euractiv.com/section/aviation/news/airlines-to-invoke-eu-freedoms-to-challenge-green-restrictions-on-short-haul-flights/> accessed: 2023-05-10.
- EUROCONTROL, 2021. Plane and train: getting the balance right. Aviation Sustainability Unit. <https://www.eurocontrol.int/sites/default/files/2021-06/eurocontrol-thin-k-paper-11-plane-and-train-right-balance.pdf>.
- European Commission, 2020. Special Eurobarometer 495: Mobility and transport. https://data.europa.eu/data/datasets/s2226_92_1_495_eng?locale=en. (Accessed 21 November 2024).
- European Environment Agency (EEA), 2021. T. Transport and Environment Report 2020: Train or Plane? EEA Report No 19/2020. ISSN 1977-8449.
- European Union Aviation Safety Agency (EASA), 2021. Study on the Societal Acceptance of Urban Air Mobility in Europe. EASA, Cologne. Available online at: <https://www.easa.europa.eu/en/full-report-study-societal-acceptance-urban-air-mobility-europe#group-downloads>. (Accessed 20 May 2025).
- European Union Aviation Safety Agency (EASA), 2022. Prototype Technical Design Specifications for Vertiports. EASA, Cologne. Available online at: <https://www.easa.europa.eu/en/prototype-technical-design-specifications-vertiports>. (Accessed 20 August 2025).
- Eurostat, 2025. Modal split of inland passenger transport. https://ec.europa.eu/eurostat/databrowser/view/TRAN_HV_PSMOD/default/table?lang=en.
- Federal Aviation Administration (FAA), 2023a. Advanced Air Mobility (AAM) implementation plan. <https://www.faa.gov/sites/aa.gov/files/AAM-I28-Implement-Plan.pdf>. (Accessed 18 August 2025).
- Federal Aviation Administration (FAA), 2023b. Urban air mobility concept of operations v2.0. U.S. Department of transportation. Federal Aviation Administration. <https://www.faa.gov/sites/aa.gov/files/Urban-Air-Mobility-Concept-of-Operations-2.0.pdf>. (Accessed 19 August 2025).
- Garuti, C.E., 2020. A set theory justification of Garuti's compatibility index. *J. Multi-Criteria Decis. Anal.* 27, 50–60. <https://doi.org/10.1002/mcda.1667>.
- Gomes, L.F.A.M., de Mattos Fernandes, J.E., de Mello, J.C.C.S., 2014. A fuzzy stochastic approach to the multicriteria selection of an aircraft for regional chartering. *J. Adv. Transp.* 48 (3), 223–237.
- Grimme, W., Jung, M., 2018. Towards more sustainability? – the development of aviation emissions from Germany between 1995 and 2016. In: *Proceedings in the 22nd Air Transport Research Society World Conference in Seoul/Korea*.
- Güntüt, C., Gökdalay, M., 2023. Aircraft selection decision support model for fleet planning of the low cost airlines. *Eskişehir Osmangazi Üniversitesi İktisadi ve İdari Bilimler Dergisi* 18 (2), 460–478.
- Gupta, S., Gupta, S., 2024. What drives customers crazy for green vehicles? A fuzzy AHP approach. *Environ. Dev. Sustain.* 26 (9), 23283–23302.
- HDI, 2022. Technical Study Electric Aviation in 2022. https://www.hdi.global/globalassets/local/international/downloads/specialty/aviation/e-flight_whitepaper.pdf. (Accessed 6 May 2023).
- Hepburn, C., Grubb, M., Neuhoof, K., Matthes, F., Tse, M., 2006. Auctioning of EU ETS phase II allowances: how and why? *Clim. Policy* 6 (1), 137–160.
- Holden, J., 2018. Uber keynote: scaling uber air. Uber Elevate Summit. <https://www.uber.com/us/en/elevate/summit/2018/>.
- Hwang, C.L., Lai, Y.J., Liu, T.Y., 1993. A new approach for multiple objective decision making. *Computers and Operational Research* 20 (8), 889–899. [https://doi.org/10.1016/0305-0548\(93\)90109-v](https://doi.org/10.1016/0305-0548(93)90109-v).
- Hwang, C.L., Yoon, K., 1981. *Multiple Attribute Decision Making: Methods and Applications*. Springer-Verlag, New York.
- IATA, 2024. Aviation supporting the global economy: vital role connecting the world. <https://www.iata.org/en/iata-repository/pressroom/fact-sheets/fact-sheet-benefits-aviation-statistics/>.
- Johnson, T., Riedel, R., Sahdev, S., 2020. To take off, flying vehicles first need places to land. McKinsey and Company. <https://www.mckinsey.com/industries/automotive-and-assembly/our-insights/to-take-off-flying-vehicles-first-need-places-to-land>. (Accessed 15 February 2023).
- Kasliwal, A., Furbush, N.J., Gawron, J.H., McBride, J.R., Wallington, T.J., de Kleine, R. D., Kim, H.C., Keoleian, G.A., 2019. Role of flying cars in sustainable mobility. *Nat. Commun.* 10, 1555.
- Keeney, R.L., Raiffa, H., 1993. *Decisions with Multiple Objectives*. ISBN 0-521-44185-4.

- Kendall, M., 1938. A new measure of rank correlation. *Biometrika* 30 (1–2), 81–93. <https://doi.org/10.2307/2332226>.
- KPMG, 2019. Getting mobility off the ground. <https://institutes.kpmg.us/manufacturing-institute/articles/2019/getting-mobility-off-the-ground.html>.
- Lyle, C., 2018. Beyond the ICAO's CORSIA: towards a more climatically effective strategy for mitigation of civil-aviation emissions. *Clim. Law* 8 (1–2), 104–127.
- Maertens, S., Grimme, W., Scheelhaase, J., 2020. ICAO's new CORSIA scheme at a glance—a milestone towards greener aviation?. In: *Aviation and Climate Change*. Routledge, pp. 117–129.
- Mallela, J., Wheeler, P., Le Bris, G., Nguyen, L.G., 2023. Urban Air Mobility: An Airport Perspective, Airport cooperative research program, ACRP research report 243. National Academies and Transportation Research Board.
- Moore, M.D., Goodrich, K., Viken, J., Smith, J., Fredericks, B., Trani, T., Barraclough, J., German, B., Patterson, M., 2013. High Speed Mobility through On-Demand Aviation. In: 2013 Aviation Technology, Integration, and Operations Conference, Los Angeles, CA, August 12–14 2013, AIAA 2013, p. 4373.
- NASA, 2021. Advanced Air Mobility (AAM) Mission. [https://www.faa.gov/sites/faa.gov/files/media/airports/2021/march/airports-mar2021-NASAAdvancedAirMobility\(AAM\)Mission.pdf](https://www.faa.gov/sites/faa.gov/files/media/airports/2021/march/airports-mar2021-NASAAdvancedAirMobility(AAM)Mission.pdf). (Accessed 18 August 2025).
- Nava, C.R., Meleo, L., Cassetta, E., Morelli, G., 2018. The impact of the EU-ETS on the aviation sector: competitive effects of abatement efforts by airlines. *Transport. Res. Pol. Pract.* 113, 20–34.
- Ozdemir, Y., Basligil, H., 2016. Aircraft selection using fuzzy ANP and the generalized choquet integral method: the Turkish airlines case. *J. Intell. Fuzzy Syst.* 31 (1), 589–600.
- Patterson, M.D., German, B.J., Moore, M.D., 2012. Performance Analysis and Design of On-Demand Electric Aircraft Concepts. In: *Performance Analysis and Design of On-Demand Electric Aircraft Concepts*. 12th AIAA Aviation Technology, Integration, and Operations (ATIO) Conference and 14th AIAA/ISSMO Multidisciplinary Analysis and Optimization Conference, Indianapolis, IN, Sept 17–19 2012, AIAA-2012, p. 5474.
- Patterson, M.D., Antcliff, K.R., Kohlman, L.W., 2018. A proposed approach to studying urban air mobility missions including an initial exploration of mission requirements. In: *Annual Forum and Technology Display* (No. NF1676L-28586).
- Polaczyk, N., Trombino, E., Wei, P., Mitici, M., 2019. A review of current technology and research in urban on-demand air mobility applications. In: 8th Biennial Autonomous VTOL Technical Meeting and 6th Annual Electric VTOL Symposium, pp. 333–343.
- Preeti, Mittal, M., Gupta, S., 2024. Prioritising the service quality factors affecting customer satisfaction in life insurance industry: an application of fuzzy AHP. *Int. J. Prod. Qual. Manag.* 42 (4), 519–535.
- Prussi, M., Lee, U., Wang, M., Malina, R., Valin, H., Taheripour, F., et al., 2021. CORSIA: the first internationally adopted approach to calculate life-cycle GHG emissions for aviation fuels. *Renew. Sustain. Energy Rev.* 150, 111398.
- Reiche, C., Goyal, R., Cohen, A., Serrao, J., Kimmel, S., Fernando, C., Shaheen, S., 2018. Urban Air Mobility Market Study. National Aeronautics and Space Administration (NASA). <https://doi.org/10.7922/G2ZS2TRG>.
- Rico, C., Heyl, B., 2025. The State of the EU's Rail Infrastructure. *Transport & Environment report* (June 2025). https://www.transportenvironment.org/uploads/files/Tracking-Europes-rail-infrastructure-REPORT_2025-06-24-072810_gqwa.pdf. (Accessed 19 August 2025).
- Riza, L., Bruehl, R., Fricke, H., Planing, P., 2024. Will air taxis extend public transportation? A scenario-based approach on user acceptance in different urban settings. *Transp. Res. Interdiscip. Perspect.* 23, 101001.
- Sama, H.R., Gupta, S., Mathew, M., Gupta, S., 2021. Comparative analysis of service quality of Big Bazaar and Spencer's: a fuzzy approach. *Int. J. Qual. Reliab. Manag.* 38 (10), 1993–2006.
- See, T.K., Gurnani, A., Lewis, K., 2004. Multi-attribute decision making using hypothetical equivalents and inequivalents. *J. Mech. Des.* 126 (6), 950–958.
- Shihab, S.A.M., Wei, P., Ramirez, D.S.J., Mesa-Arango, R., Bloebaum, C., 2019. By schedule or on demand A hybrid operation concept for urban air mobility. AIAA Aviation 2019 Forum. AIAA Aviation 2019 Forum, Dallas, TX, June 17.
- SRIA (Strategic Research and Innovation Agenda), 2020. Digital european sky. <https://www.sesarju.eu/sites/default/files/documents/reports/SRIA%20Final.pdf>.
- Straubinger, A., Rothfeld, R., Shamiyeh, M., Büchter, K.D., Kaiser, J., Plötner, K.O., 2020. An overview of current research and developments in urban air mobility—setting the scene for UAM introduction. *J. Air Transport. Manag.* 87, 101852.
- Sun, J., Olive, X., Strohmeier, M., 2022. Environmental footprint of private and business jets. *Engineering Proceedings* 28 (1), 13.
- Sun, X., Gollnick, V., Stumpf, E., 2011. Robustness consideration in multi-criteria decision making to an aircraft selection problem. *J. Multi-Criteria Decis. Anal.* 18 (1–2), 55–64.
- The Wall Street Journal, 2025. Joby takes key step toward commercial service with first airport-to-airport flight. https://www.wsj.com/business/airlines/joby-takes-key-step-toward-commercial-service-with-first-airport-to-airport-flight-44cdbc2?utm_source=chatgpt.com. (Accessed 22 August 2025).
- Vieira, D.R., Silva, D., Bravo, A., 2019. Electric VTOL aircraft: the future of urban mobility (background, advantages and challenges). *International Journal of Sustainable Aviation* 5, 2.
- Volocopter, 2024. Volocopter Conducts First Evtol Flight at the Palace of Versailles. Volocopter GmbH. Available at: <https://www.volocopter.com/en/newsroom/vc-flight-versailles-palace>. (Accessed 18 August 2025).
- Wadud, Z., 2017. Fully automated vehicles: a cost of ownership analysis to inform early adoption. *Transport. Res. Pol. Pract.* 101, 163–176.
- Wang, Y., Yuan, K., Zhu, M., Li, S., 2023. A time-and-space-network-based green fleet planning model and its application for a hub-and-spoke network. *Sustainability* 15 (7), 5832.
- Wang, Y.M., Chin, K.S., 2011. Fuzzy analytic hierarchy process: a logarithmic fuzzy preference programming methodology. *Int. J. Approx. Reason.* 52 (4), 541–553. <https://doi.org/10.1016/j.ijar.2010.12.004>.
- Witlox, F., Zwanikken, T., Jehee, L., Donners, B., Veeneman, W., 2022. Changing tracks: identifying and tackling bottlenecks in European rail passenger transport. *European Transport Research Review* 14 (1), 7.
- Yedavalli, P., Mooberry, J., 2019. An assessment of public perception of Urban Air Mobility (UAM). https://storage.googleapis.com/blueprint/AirbusUTM_Full_CommunityPerceptionStudy.pdf.
- Yeh, C.H., Chang, Y.H., 2009. Modeling subjective evaluation for fuzzy group multicriteria decision making. *Eur. J. Oper. Res.* 194 (2), 464–473.
- Yoon, K., 1987. A reconciliation among discrete compromise situations. *J. Oper. Res. Soc.* 38 (3), 277–286. <https://doi.org/10.1057/jors.1987.44.S2CID.121379674>.